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BURNED FOREST LAND COVER CLASSIFICATION FOR CO₂ EMISSIONS CALCULATION

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Anton Donle: Burned Forest Land Cover Classification for CO2 Emissions Calculation
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Abstract

Forest ecosystems are increasingly threatened by climate change and human activity. Portugal is a country that has suffered from especially severe and frequent fires in recent years. This research develops a novel method to assess the damage that a forest fire caused, by making a more precise estimation of the forest land cover, and the carbon emissions produced by the fire. The research consists of three main steps. Based on remote sensing the burned area after a fire is detected. A forest land cover map is generated by combining satellite images with the Portuguese forest inventory, indicating the dominant tree species with a resolution of 10 meters. Lastly, the pre-fire biomass in the forest, and the carbon emissions caused by the fire are calculated on the basis of the forest composition. The research is carried out on the largest fire of 2024 in Portugal, located in the parish of Reriz e Gafanhão. The satellite images from the Sentinel-2 mission are used.

The originality of this thesis has been checked using the Turnitin OriginalityCheck service.

Keywords: Remote Sensing, Forest Fire, Carbon Emissions, Burned Area Detection

Author's Declaration of Originality

I hereby certify that I am the sole author of this thesis. All the used materials, references to the literature and the work of others have been referred to. This thesis has not been presented for examination anywhere else.

Author: Anton Donle

A handwritten signature in black ink, appearing to read 'A. Donle', written in a cursive style.

April 28, 2026

Declaration of use of Large Language Models ("AI")

I have utilised the following LLM tools in my thesis.

- Gemini Flash 2.5 (built into google colab) and notebooklm
- Gemini 3.1, Gemini 3.5
- Chatgpt Versions: 4.1, 5.2, 5-mini (<https://chatgpt.com/>)

Description of how they were used: In this work, chatGPT was used to navigate websites in Portuguese. It was used to translate and browse data related to the Portuguese national forest inventory and to identify relevant sections in a report in Portuguese. Moreover, ChatGPT was used to format tables that I generated in Jupyter Notebooks into a suitable Overleaf format to avoid tedious and repetitive manual labor. In the tables with Portuguese class names (from the NFI), chatGPT was used to translate them to English according to a translation convention that I gave as input. Gemini is integrated into the google Colab environment. It was used to solve bugs in the python code. It was also used to get inspiration on how to approach programming problems. NotebookLM is a tool used to gain an overview of papers. It uses a version of the Gemini LLM from Google. It was used to identify the part of the paper relevant for the problem I was working on. For example, after I finished reading a paper and taking notes, I wanted to find out what resolution or exactly how many samples were used, I asked it to highlight the relevant parts of the paper where this was discussed. Gemini was used to do text enhancements (help to identify missing points or parentheses).

I acknowledge that I am fully responsible for the entire content of my thesis, including the parts generated by AI, and accept accountability for any violations of ethical standards in publications.

List of Abbreviations

AGB Above Ground Biomass

AI Artificial Intelligence

NDVI Normalized Difference Vegetation Index

NBR Normalized Burn Ratio

dNBR difference Normalized Burn Ratio

SNAP Sentinel Application Platform

SWIR Short Wave Infra Red

NIR Near Infra Red

NFI National Forest Inventory

RQ Research Question

Mg/ha Megagram per Hectare

Mt Megaton

List of Translations From Portuguese NFI

Below is a list that contains the original NFI6 names in Portuguese and the English names that they were translated to:

Pinheiro-bravo Maritime Pine

Eucaliptos Eucalyptus

Carvalhos (Calvos) Oak

Castanheiro Chestnut

Outras folhosas Other Deciduous

Outras resinosas Other Conifers

Matos Shrubland

Improdutivos Unproductive Area

Urbano Urban Area

Povoamento em pé Standing forest stands

Cortes únicos Single harvest stands

Povoamentos em regeneração Regenerating stands

Povoamentos ardidos Burned stands

usosolo Land Use

ocupprinci Primary land cover

ocupsecund Secondary land cover

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Introduction

Forests are a crucial part of the ecosystem of our planet. Recently, climate change and human activity have been threatening forests around the world. Mediterranean climates, such as Southern Europe, in particular, are experiencing hotter and drier summers, leading to more frequent and more severe forest fires. In recent years, Portugal has suffered several extreme fire events. Forest fires cause the loss of trees and with that their ability to perform photosynthesis. The carbon that was stored in trees is released into the atmosphere in the form of CO₂ emissions.

In 2017, there were 21,000 forest fires in Portugal, destroying a total of 530,630 hectares of forest (European Commission: Joint Research Centre, 2018). In the Pedrógão Grande fire complex, there were also 66 human casualties (Rodrigues et al., 2022). This research aims to produce a framework for analyzing the climate impact of forest fires using open source data. For this, a case study of a forest fire in Portugal in 2024 is performed. The fires of 2024 were less severe than the 2017 fires, but due to their recency, the damage they caused has not yet been thoroughly investigated and quantified. The method developed in this research aims to establish and implement a pipeline for the rapid assessment of post-fire environmental impacts. Understanding the impact and extent of the damage caused by a forest fire is important both for preventive and restorative forest management. Accurate estimates of biomass loss enable better assessments of both ecological and economic damage caused by the fire. This information is relevant for Portuguese forest authorities, for example, for supporting climate impact mitigation efforts. This research aims to estimate the types of vegetation that were burned in the forest fire and to quantify their prefire biomass. These factors serve as a basis for estimating how much CO₂ was emitted during the forest fire.

For calculating precisely how much carbon is stored in a tree, measurements at the individual tree level are necessary. Those measurements would include parameters such as diameter at breast height and the height of trees (Ferreira et al., 2025) (Lu et al., 2016). Instead of manually measuring and calculating the biomass for each tree separately, approaches have been developed that estimate the biomass using field measurements and remote sensing (Gao & Zhang, 2021) (Moradi et al., 2022). Direct measurements are a destructive method in which trees are cut down and the biomass is inferred from their volume. This method is expensive and time consuming and it is generally accepted that it is not suitable for large scale measurements (Moradi et al., 2022). Instead, often only the exact amount of carbon is determined for samples, which are combined with satellite images to predict biomass on a larger scale with machine learning models (Lu et al., 2016), (Moradi et al., 2022), (Georgopoulos et al., 2022).

A crucial first step in assessing the carbon in a forest is to identify which trees the forest consists of and how they are distributed over the area. The reports on forest fires by the European Commission indicate the type of forest that has been burned, but do not provide in depth information on the species of trees that have been burned or the composition of these forests (European Commission: Joint Research Centre, 2025).

The Portuguese National Forest Inventory (NFI) (Uva et al., 2021) provides information on forest coverage, but the information is sparse and collected only in long time intervals of 10 years (ICNF, 2019b). In Portugal, this information is represented as discrete points, 500 meters spaced apart. The NFI points provide an overview of what species are present, but due to the low density of this information and its discrete nature, this is not enough to make a claim about the distribution of the forest.

A land cover map describes the surface of the Earth. Land cover maps of forests offer a structured, continuous and visual representation of the species of the vegetation present at each location. Depending on the resolution and type of the map, this can give information about individual trees or groups of trees of one species. A land cover map of the vegetation types in a forest is useful because it provides a more detailed description of the forest, instead of labeling the whole forest as, for example, a "broadleaf forest". A common approach to derive a higher resolution land cover map of forests is to combine the unlabeled Sentinel-2 images as a feature space with national forest inventory data as labels using machine learning algorithms (Moraes et al., 2025), (Blickensdörfer et al., 2024), (Hemmerling et al., 2021). This approach overcomes the low information density of the forest inventory data and aims to provide an accurate label for each of the pixels in the satellite image.

Generally, remote sensing can be used both for creating higher resolution land cover maps (Isbaex & Coelho, 2021), (Pandit et al., 2018) and to estimate the biomass of a forest (Lu et al., 2016), (Chen et al., 2019), when combined with labels in the form of forest inventories or field measurements. The Copernicus Sentinel-2 images are suitable for this. The Sentinel-2 mission consists of twin satellites, with a frequent revisit time of approximately 6 days. They provide spectral data on different bands that can be used to calculate further indices.

The aim of this research is to quantify more precisely how much CO₂ was released during a forest fire, by calculating the fire emissions on the level of individual vegetation types, rather than relying on a broad average for the entire forest area. Hence, the question (RQ1) this research aims to answer is:

To what extent can a more precise estimation of a forest composition, using both forest inventory data and Sentinel-2 satellite images, enhance the accuracy of the estimation of CO₂ emissions produced by a forest fire?

To calculate the CO₂ emissions that the burning of specific types of vegetation produced, it is necessary to understand the composition of the forest before the fire by estimating what percentage of the forest was occupied by what kind of vegetation. To address this, the following sub-research question (RQ2) is proposed:

Using Sentinel-2 images and forest inventory data, to what extent can machine learning algorithms generate a more precise land cover map of the forest before the fire, when compared to using only forest inventory data?

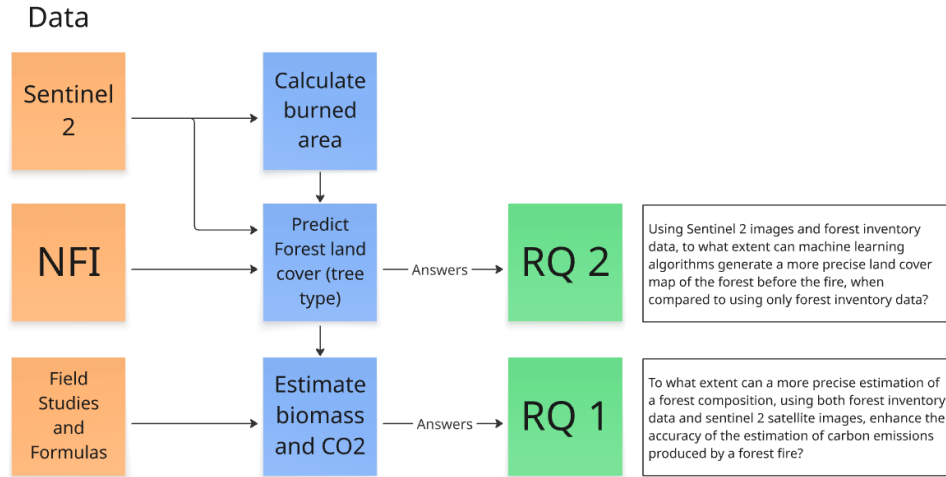


Figure 1: Workflow and Research Questions

Figure 1 gives an overview about what data is used and what methodology is applied to answer the formulated research questions. In Section 1 an overview of the existing approaches for identifying burned areas, predicting forest land cover maps, and estimating biomass loss is given. In Section 2 the data sources, tools, methods, and algorithms are explained. First, the exact area of the largest forest fire in Portugal in 2024 is determined based on Sentinel-2 images taken before and after the forest fire. The Portuguese NFI (IFN6) and Sentinel-2 images are used to train a machine learning classifier to create a land cover map of the vegetation types in the burned forest. Based on the land cover map, a formula will be used to estimate the amount of biomass loss and carbon emissions, due to the fire. Section 3 gives an overview of the results. In Section 4 the land cover map is validated against the CORINE forest type map. Section 5 discusses the findings and limitations of this work and puts them into context. Lastly, Section 6 concludes and summarizes the work.

1 Literature Review

Quantifying CO₂ emissions caused by a forest fire is an interdisciplinary challenge that requires building on findings from different fields. This review serves as the theoretical foundation and is structured into three main parts. First, it explores remote sensing approaches to precisely identify the burned area in a forest fire. Secondly, it examines methodologies for combining Sentinel-2 images with national forest inventories to create a land cover map of the vegetation species in the forest. Lastly, building on ecological principles, it analyses the parameters and equations required to calculate pre-fire biomass and CO₂ emissions from a fire.

1.1 Identifying the Burned Area in a Forest

According to White et al. (1996) the chlorophyll in the leaves of trees strongly reflects near infrared (NIR) light. Burned trees lose that property and reflect less NIR light. Their research also showed that a burned plant shows higher reflectiveness for shortwave infrared (SWIR) light than before. Using this principle Key and Benson (2006) formalized the normalized burn ratio (NBR), which is used to indicate burned areas. The NBR is calculated using the NIR band and the SWIR band from a satellite image. Calculating the difference in NBR before and after the fire gives an even better indication of the burned area in a forest fire. A large NBR indicates healthy vegetation; hence, the greater the (positive) difference in NBR (dNBR) before and after the fire, the more the vegetation was damaged. Parks et al. (2014) introduced the relativized burn ratio (RBR) as a new measurement of the severity of a fire. The RBR is calculated by normalizing the dNBR. They showed that it outperforms the dNBR in classifying the severity of a forest fire, when using Landsat satellite images. Arisanty et al. (2022) investigated the effectiveness of using the normalized burn ratio (NBR), the difference normalized burn ratio (dNBR) and the relativized burn ratio (RBR) at detecting the burned area and assessing the severity of a forest fire using Sentinel-2 images. In Sentinel-2 images, the band 8 corresponds to NIR light, and the band 12 corresponds to SWIR light. Their research showed that the RBR was the most accurate of the three ratios in assessing fire severity and detecting burned areas. Fotakidis et al. (2023) compared the three ratios for continuous burned area mapping and found that the NBR performs significantly worse than the other two. They also found that the dNBR and RBR perform similarly well, with the dNBR performing slightly better.

Based on previous research for the binary classification of burned areas using Sentinel-2 data, both RBR and dNBR are reasonable choices. They were both shown to have good performance. This research will use dNBR to detect the area burned by the forest fire.

Arisanty et al. (2022) present a table from the United States Geological Survey USGS with dNBR thresholds for fire severity. They classify areas with dNBR values smaller than 0.1 as unburned. Areas with values larger than 0.1 are classified as a low severity burn and areas with values > 0.27 are classified as a moderate-low severity burn. This research will aim to identify all burned areas. Therefore, 0.1 will be used as the threshold to identify the burned area.

1.2 Portuguese National Forest Inventory in Forest Land Cover

National Forest Inventories are systematic measuring programs by many countries to analyze and report data about the extent, composition, and trends of national forests (Tomppo et al., 2010). The Portuguese National Forest Inventory (NFI) provides valuable information about forests in Portugal in the form of labeled points. It exists since 1965 and is repeated every 10 years (ICNF, 2019a). The most recent version NFI6 is from 2015. ICNF (2019b) states that the forests in Portugal can be grouped into four major groups according to their structure and function. The NFI6 consists of roughly 360 thousand points that are spread out in a

grid over the whole country. The points have a distance of 500 meters from each other (Uva et al., 2021). Each of these points indicates the land use class (such as forest, urban area or shrubland), the primary and secondary land use, and in case the point indicates a forest, it also contains information about its size and density. However, the NFI does not provide information about how land cover extends in relation to the indicated point (e.g., its boundaries, shape, or orientation). Moraes et al. (2025) have discussed that due to the lack of spatial continuity, the relatively large distance between points, and the infrequent updates, the NFI alone is not enough to create a detailed and up to date land cover map of Portugal. It is not possible to use the NFI to identify specific trees or specific groups of trees. Kovárník and Janová (2025) point out that the NFI faces challenges such as high costs and the limited update frequency, while Ganz et al. (2020) discuss that data derived from forest inventories are often out of date or oversimplified. Moraes et al. (2025) identify that the NFI points in Portugal provide reliable information only in a small area around the position of the points.

The NFI provides sparse, discrete points. In order to create a continuous high resolution map, in this research, the NFI will be combined with Sentinel-2 images, using a machine learning approach.

1.3 Sentinel-2 Bands and Indices as Features for Land Cover Prediction

The Sentinel-2 mission provides high-resolution, multispectral imagery. The data is suitable for land surface observation and land cover prediction (Phiri et al., 2020). It contains 13 different spectral bands. The range of the spectral bands is from 443 nm (reflects coastal aerosol) to 2190 nm (reflects short wave infra red light). The bands have a resolution between 10m-60m. Spectral indices, such as the normalized difference vegetation index (NDVI), can be calculated based on the bands. Phiri et al. (2020) point out that Sentinel-2 data can be used to predict land cover in different areas such as agricultural, urban, and forest regions. Different researchers use a different subset of bands and indices, depending on the type of land cover they predict.

Blickensdörfer et al. (2024) and Hemmerling et al. (2021) classify the dominant tree species in German forests. They use the spectral bands in the visible range: 2 (blue), 3 (green) and 4 (red), the bands 5, 6 and 7 which are used for classifying vegetation, and band 8 (near infra red light) and the short wave infra red light bands: 11 and 12. They also used the normalized difference vegetation index (NDVI). Moraes et al. (2025) used the same subset, and in addition to that also band 10 (cloud detection) and the indices: normalized burn ratio (NBR), normalized difference water index (NDWI), normalized difference burn index (NDBI), since they aimed to produce a land cover map of the whole country and not just forest regions. The properties and descriptions of the bands are based on Table 1 from Kaplan and Avdan (2017) and the Sentinelhub website about Sentinel-2 bands. The NDVI as formulated by Tucker (1979) indicates leaves that actively perform photosynthesis. Hence the larger the NDVI, the more living vegetation is present. Ganz et al. (2020) used only the NDVI in combination with

airborne LiDAR sensors to perform binary classification of whether a pixel contained forests or not. To distinguish between crops, forests, construction and water Kovárník and Janová (2025) used bands: 2, 3, 4, 8, 11, 12 and the NDVI. The Table 1 gives an overview of the Sentinel-2 bands and indices used in the aforementioned research.

In this research, the Portuguese NFI will be combined with Sentinel-2 images to predict a more precise forest composition of the burned area in a forest. Since this is closest to the approaches of Hemmerling et al. (2021) and Ganz et al. (2020), the same subset of bands and indices will be used as the feature space for this work. The main difference is that they worked with the German NFI, which provides a significantly different data format than the Portuguese NFI. Moraes et al. (2025) produced a land cover map for the entire continental Portugal area with the use of the Portuguese NFI. They did not put special emphasis on forests and other types of vegetation. This research will only focus on the land cover of one specific forest and ignore areas that have not been burned in the fire.

1.3.1 Data preparation of Sentinel-2 Images: Resolution and Pixel Size

The different bands in Sentinel-2 images have a resolution between 10 m x 10 m and 60 m x 60 m. For dealing with pixels of different sizes, the common strategy is to resample all pixels at a resolution of 10 m x 10 m (Blickensdörfer et al., 2024), (Moraes et al., 2025), (Hemmerling et al., 2021), (Kovárník & Janová, 2025).

A resolution of 10 m is too low to detect single trees. However, it can be used to identify the majority of trees in a pixel. In order to achieve higher spatial resolutions Ganz et al. (2020) combine aerial sensors with Sentinel-2 data. Other approaches such as airborne LiDAR sensors and spectrometry have been used to acquire very high spatial resolution and predict land cover for pixels of 1 m x 1 m (Koetz et al., 2008). The satellite images are either collected as a time series, and pixel values are averaged over a certain time period, or specific, discrete images are used as features. Moraes et al. (2025) sampled the satellite images for one year and calculated the average values per month, processing them as a time series.

In this research, the limitation of 10 m x 10 m of Sentinel-2 will not be overcome, and the focus will be to predict the majority of trees in each pixel as accurately as possible. Instead of making use of a time series over a longer period of time, a pre-fire and a post-fire image are used.

1.3.2 NFI Data as Labels for Machine Learning Approaches to Land Cover Prediction

Sentinel-2 data provides a lot of information, yet pixels are not labeled, making it difficult to interpret the data or extract meaningful information about land cover (Moraes et al., 2025).

Table 1: Overview About Papers, Sentinel-2 Bands and Indices, Classes, Labels, and Algorithms

Citation	Sentinel-2 Bands and Indices	Classes	Labels	Algorithms
Blickensdörfer et al. (2024)	Bands 2, 3, 4, 5, 6, 7, 8, 11, 12, NDVI	Forest land cover: 11 tree species	German NFI	Random Forest Classifier
Hemmerling et al. (2021)	Bands 2, 3, 4, 5, 6, 7, 8, 11, 12, NDVI	Forest land cover: 17 tree species	German NFI	Random Forest Classifier
Ganz et al. (2020)	NDVI	Trees vs. no trees	validation against German NFI	Gaussian Mixture Model
Kovárník and Janová (2025)	Bands 2, 3, 4, 8, 11, 12; NDVI	4 land cover classes: Crop, Forest, Construction, Water	manual labels; validation with Czech NFI	Random Forest, CART, SVM, Naive Bayes (NB)
Moraes et al. (2025)	Bands 2, 3, 4, 5, 6, 7, 8, 10, 11, 12; NDVI, NBR, NDWI, NDBI, NDMIR	19 land cover classes (urban, winter crops, eucalyptus, shrubland, . . .)	Portugal NFI	CNN, masked autoencoder-like approach

Using NFI as labels for a machine learning algorithm (Blickensdörfer et al., 2024), (Hemmerling et al., 2021), (Moraes et al., 2025) or for validation of its performance (Kovárník & Janová, 2025), (Ganz et al., 2020) has been done by different researchers in the past. Blickensdörfer et al. (2024) used the German NFI labels, in combination with Sentinel-1 and Sentinel-2 data

(as a time series) to distinguish between 11 tree classes in German forests. They report a high accuracy with the top five dominant tree species and difficulties in identifying less frequent species and mixed forests. The classes they are predicting are: Spruce, Pine, Douglas fir, Fir, Larch, Beech, Oak, Birch, Alder, other deciduous trees with high life span (ODH) and other deciduous trees with low life span (ODL). In a similar manner, Moraes et al. (2025) use the Portuguese NFI as labels, after filtering to increase the label quality. Hemmerling et al. (2021) also use the German NFI in combination with Sentinel-2 images (time series) to predict 17 tree species in the state of Brandenburg. Ganz et al. (2020) make a binary classification to distinguish forest from non-forest on a spatial resolution of 1 meter. They use a multi sensory approach, combining Sentinel-2 data with aerial images. Kovárník and Janová (2025) use manually labeled points and then validate their results by comparing them to the Czech NFI.

This research will use Portuguese NFI as labels and perform similar processing steps as Moraes et al. (2025) to ensure their quality and to increase the number of labels available.

1.3.3 Algorithms for Land Cover Prediction

Different researchers have used Random Forest (RF) (Blickensdörfer et al., 2024), (Hemmerling et al., 2021), (Kovárník & Janová, 2025) or Support Vector Machine (SVM) (Koetz et al., 2008), (Kovárník & Janová, 2025) classifiers to create a land cover map. These classifiers treat each pixel individually. They are easier to train with a low number of annotated pixels, but they do not inherently reflect the spatial relationship between pixels. Moraes et al. (2025) use a weakly supervised method and a Convolutional Neural Network (CNN). The advantage of a CNN algorithm is that it can understand more intricate patterns and spatial relationships between pixels. However, it usually relies on fully pixel-level annotated images.

In this research, SVM classifier and Random Forest classifier will be used and their performance will be compared.

1.3.4 Other Forest Land Cover Datasets

In addition to national forest inventories that provide information on exact tree species at a low resolution, there are datasets that indicate forest cover at a high resolution for Europe or worldwide. However, none of the existing resources indicates the exact vegetation species on a sufficiently high resolution for more precise CO₂ emission calculations. The ESA WorldCover 10 m dataset (Zanaga et al., 2022) provides land cover information at a 10 m resolution, but only indicates "Forest cover" without giving more detailed information on the type of forest. Similarly, the Copernicus High Resolution Layer (HRL) Forest Type dataset (European Union's Copernicus Land Monitoring Service, 2021) provides a forest type dataset, which contains information about the type of forest at a resolution of 10 m. More specifically, it indicates whether a certain pixel contains coniferous forest, broadleaf forest, or not forest. These datasets indicate the type of forest at a high resolution, yet, they do not contain information on the specific

kind of tree. Neither of those maps alone is sufficient to produce a detailed, tree specific map of forests in Portugal. More information is needed to infer the distribution of specific vegetation types in a forest.

1.4 Estimating CO₂ Emissions Caused by a Fire Through Forest Biomass

The biomass of a forest describes how much carbon is stored in the trees (Lu, 2006). This is relevant and strongly related to how much CO₂ they release when burned in a fire. The biomass in forests consists of two parts: the above ground biomass (AGB), such as leaves, stem, branches, and the below ground living mass, such as the roots and dead fine and coarse litter in the soil (Lu et al., 2016), (Lu, 2006). Due to difficulties in measuring the below ground biomass, most research only takes into account the above ground biomass (Lu, 2006). According to Massey et al. (2023) the AGB of the forest is expressed as the dry weight of live trees per unit of area. A common unit for this is megagram per hectare (Mg/ha). A megagram is equivalent to one million gram, hence the same as a metric ton. In this research, the unit and notation Mg/ha will be used.

The biomass of trees can be estimated and calculated in various ways. Different approaches use different types of data and have varying degrees of precision. The most accurate way to calculate biomass is through (destructive) field measurements (Alegria, 2023). This is often described as too resource intensive on a larger scale (Moradi et al., 2022). Based on values extracted from destructive methods, allometric formulas can be calculated which only require features that can be measured with non-destructive methods; hence, the measuring becomes easier and less intrusive. As a more scalable solution, remote sensing approaches have been used, which use field studies as ground truth.

1.4.1 Existing Biomass Datasets

Global datasets that describe forest biomass tend to have a very low resolution, averaging biomass over large areas. Santoro et al. (2021) discuss that global AGB datasets with a low resolution of ≥ 500 m do not capture fine scale variability in biomass. Their research produces a forest AGB map with a pixel size of one hectare (100 x 100 m). Generally, research and datasets that calculate the lost or burned biomass after a forest fire on a larger scale tend to estimate the percentage of biomass burned. They use an estimate of fuel type depending on the vegetation. French et al. (2011) discuss that traditionally these estimates are based on three parameters: area burned, fuel load, and proportion of biomass consumed. These calculations tend to use average values for forest types or larger areas.

Wiedinmyer et al. (2011) provide global estimates of particle emissions from burned biomass, for wildfires, among other types of fire, on a resolution of 1 km. Another dataset that assesses fire emissions is the GFED (Global Fire Emissions Database) (van der Werf et al., 2025). The dataset is claimed to be mostly suitable for large-scale atmospheric studies. The

spatial resolution is 0.25 degrees, which corresponds to a pixel size of roughly 26 - 27 km. Another relevant dataset is the EU dataset: Forest biomass map of EU27 for 2020 (Avitabile, 2022). This provides information about the biomass on an area of 100 m x 100 m. Each pixel covers an area of 10,000 m^2 . For a precise estimation of landscapes, the low resolution or an average over the whole forest leads to large inaccuracies, which this research will aim to overcome. This research aims to estimate biomass on a high resolution (10 m) for the study area

1.4.2 Biomass in Portuguese NFI6

The Portuguese NFI contains information on the biomass of forests (ICNF, 2019b). However, this information is not linked to their geo-spatial location. It indicates for the different tree species in Portugal how much area they cover (in hectares), how much biomass they contain (in megatons), and their biomass equivalent of CO₂ (in megatons). A megaton is one million tons (1,000,000 tons). For the species relevant for this work, this information is extracted and visualized in Table 2.

Table 2: Reported Area, Biomass, and Stored Carbon per Tree Species in Portugal, From ICNF (2019b)

Species	Area (ha)	Biomass (Mt)	CO ₂ equivalent (Mt)
Maritime Pine	615,000	46,1	84,6
Eucalyptus	766,000	34,9	64,0

Tomppo et al. (2010) discuss the consistency between forest inventories internationally. They point out that the AGB reported in NFIs differ between different countries, because the estimates are mostly based on the dbh and the NFIs use a different minimum dbh threshold for trees they record individually. These thresholds range from 0 to 12 cm. The Portuguese NFI 6 uses a threshold of 7.5 cm for most species (and 5 cm for eucalyptus) for individual tree measurements. Trees with a dbh lower than that are not measured individually. Their biomass is estimated using tree height and species averages (ICNF, 2019c).

1.4.3 Estimating Biomass Using Field Studies

Field studies are the most direct and accurate way to calculate the biomass of trees in a forest (Lu et al., 2016). Lu et al. (2016) provide an overview of different sampling methods and their characteristics. In destructive sampling, trees are cut down and dried and their weight is measured. This approach was initially used and described by Klinge et al. (1975). Lu et al. (2016) also describe allometric models for calculating the tree biomass. In this type of field measurements, features such as the diameter at breast height (dbh), tree height, and wood density (constant, depending on the tree species) are determined. Based on these factors, linear or nonlinear statistical models are used to calculate the biomass in the different components of a

specific tree (Alegria, 2023). The allometric method is described as less accurate, because those factors alone may not capture variabilities in the local climate, soil conditions, and land use history (Alegria, 2023). Moreover, Peichl and Arain (2007) researched that allometric formulas work well within the forest age-sequence. They suggested that applying allometric equations without considering the age of the forest stand can lead to large errors in biomass estimation.

Moradi et al. (2022), Alegria (2023) and Motlagh et al. (2018) describe obtaining field measurements as commonly resource-intensive and time intensive. This renders it not suitable for large scale measurements. This research will not perform field studies and instead relies on values reported by other studies for determining the biomass in different vegetation types.

Biomass in Portuguese Shrubland To estimate biomass in shrublands, only the percentage of ground cover and the average height of the shrubs are necessary (Alegria, 2023), (Viana et al., 2012). Hence, the biomass of shrubs can be expressed as a function of ground cover (crown closure) and mean height of the shrubs.

Viana et al. (2012) found that the best equation for predicting shrubland biomass was Equation 1:

$$AGB_{shrubland} = (0.0258 * GC * hs)^{0.754}, \quad (1)$$

The formula consists of a multiplication of a constant, CC (crown cover as %), hs (mean shrub height in cm), and a constant power of 0.754. The result is AGB in Mg/ha. The values are based on field studies they conducted in Center-North Portugal. They collected 20 (destructive) field samples of shrubs, with a plot size of 10 m². The measured biomass was used to adjust the parameters in the regression model.

Viana et al. (2012) measured the ground cover and mean height of shrubs in 102 field plots with an area of 500 m². Based on the same formula and 30 shrubland sample plots in central Portugal with a mean ground cover of 43%, a mean height of 117.7 cm Alegria (2023) determined the mean AGB as 16.4 Mg/ha. They did not provide information on the mean height of their samples. Since in this research only the area and not the height of Shrubs is estimated, the mean AGB value will be used. Since the measurements by Alegria (2023) stem from the north of Portugal, they are considered to be more accurate and representative of the local vegetation than global averages.

Biomass of Portuguese Eucalyptus Stands Alegria (2023) collected 30 circular field sample plots with a size of 100 m² for Eucalyptus trees. They recorded the number of trees per hectare, the mean diameter, mean height, dominant diameter and dominant height and used them to calculate the AGB. The AGB was calculated as a sum of the stem under bark, the bark, the branches and the leaves. The individual components are calculated based on the recorded features. The minimum ABG was 27.7 Mg/ha, the maximum 169 Mg/ha and the mean AGB

78.7 Mg/ha with a standard deviation of 35 Mg/ha. The number of trees per hectare ranged from 800 to 4500 with a mean of 1923.

1.4.4 Estimating Biomass Using Remote Sensing

In remote sensing approaches, field measurements are used as ground truth to train machine learning algorithms. This has the advantage that the information from the field studies can be scaled to a larger region. It is a common approach to use the field study as true (labeled) data and train machine learning algorithms on remote sensing data to predict the above ground biomass over a whole forest region. Fassnacht et al. (2021) have described that only using Sentinel-2 images without airborne sensors such as LiDAR introduces the challenge of estimating AGB without information about the vertical structure of the forest. Structural information such as canopy height or volume improve accuracy of biomass estimations when compared to just using passive optical sensors, since they may suffer from signal saturation in dense forests (Tian et al., 2023), (Lu, 2006). However, since no field data could be acquired, this research does not use a machine learning approach for inferring the biomass. Instead it relies on an allometric approach.

1.4.5 Estimating CO₂ Emissions From Prefire Biomass

Quantifying CO₂ emissions from biomass in a forest follows chemical principles and has been extensively researched. The Intergovernmental Panel on Climate Change (Eggleston et al., 2006) established that when carbon is converted to carbon dioxide (CO₂), based on the molecular weights, the carbon weight can be converted to CO₂ weight with a ratio of 44/12 (Chapter 2). In Chapter 4, they point out that the carbon fraction (proportion of carbon in the biomass) has a default value of 0.47 and, depending on the forest structure, may range between 0.43 and 0.55 (Eggleston et al., 2006). Multiplying the carbon fraction (0.47) with the carbon to CO₂ ratio (44/12) results in a factor of 1: 1.72 between biomass and CO₂. They further point out that for different vegetation types on average different combustion factors apply. The combustion factor describes the proportion of prefire biomass consumed in a fire. For Eucalyptus forests, they report a value of 0.63, for other temperate forests 0.45 and for shrublands (general) 0.95.

In the Portuguese NFI, it is reported how much biomass of each tree species is present in Portugal. Moreover, the biomass equivalent in CO₂ that each species stores is reported (ICNF, 2019b). For Maritime Pine, 46.1 Mt of biomass is reported to store an equivalent of 84.6 Mt of CO₂, which corresponds to a factor of 1.835 and Eucalyptus trees with a biomass of 34.9 Mt are reported to have a CO₂ equivalent of 64 Mt, which corresponds to a factor of 1.833.

1.5 Literature Gap and Contribution

This research aims to develop a framework that can be reused to quickly assess the damage a forest fire caused, using publicly available data and a reproducible workflow. The research aims

to be reproducible, by using publicly available datasets and free software. The pipeline relies on Sentinel-2 images and forest inventory data as input. It produces an intermediate land cover map and the final result is an estimate of the CO₂ emissions in Mg.

Most of the studies in this area either produce a land cover map based on Sentinel images, or aim to estimate CO₂ emissions based on field studies. The gap that this research aims to bridge is integrating the two separate approaches for land cover mapping and carbon emissions into one workflow. This is achieved by building on the findings and reusing methodologies that have shown prior success for the individual sub tasks.

A land cover map is produced, which serves as input for an allometric formula that quantifies the fire emissions. The methodology is applied to the *Reriz e Gafanhão* fire of 2024. An estimate of the forest vegetation distribution and of the fire emissions is produced.

Furthermore, from a scientific perspective, this research investigates the efficacy of using a post fire image (compared to using a pre-fire image) for identifying the tree species and tests different parameters in the machine learning approach.

2 Methodology

This work establishes a pipeline, for post fire CO₂ emissions calculation, which is applied on the largest fire of 2024 in Portugal. The fire occurred in the period between 15.9.2024 and 20.9.2024 in *Reriz e Gafanhão*. In that epoch, there were several severe fire events in Portugal. According to the European Commission (European Commission: Joint Research Centre, 2025), in 2024, forest fires in Portugal burned 147,461 hectares of land, more than in the previous six years. Furthermore, they report that in 2024 15,357 hectares of broadleaf forest, 10,799 hectares of coniferous forest and 10,327 hectares of mixed forests were burned in Portugal. Since the 2024 data has only recently been published, not much work has been done on those fires yet. Therefore, understanding the environmental impact of these fires presents a novel and relevant challenge.

2.1 Study Area

In 2024 approximately 90% of the fires in Portugal occurred in September. The largest of these fires was in the parish of *Reriz e Gafanhão*. Figure 2 from the EFFIS report visualizes the fire events and their size. This study will focus only on the fire in *Reriz e Gafanhão*. It burned 35,523 hectares of forest, making it the largest forest fire in 2024 in Portugal. For this purpose, the Sentinel-2 tile: 29TNF will be used (see Figure 3) (European Space Agency, 2024). It fully contains the area of the fire. Two satellite images are used: one before and one after the fire. The image before the fire was taken on 16.08.2024 and the image after the fire on 30.09.2024. They were sampled based on the availability of cloud-free Sentinel-2 images (no cloud cover in relevant areas). The image selection was constrained by atmospheric conditions,

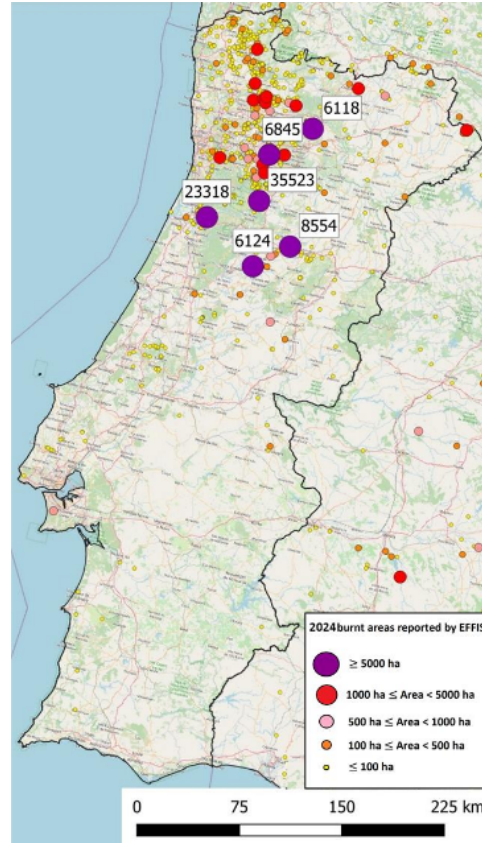


Figure 2: Burned Areas in Portugal in 2024, Reported by EFFIS, Reprinted From (European Commission: Joint Research Centre, 2025)

since optical satellites like Sentinel-2 cannot penetrate cloud cover. This research contains modified Copernicus Sentinel data [2024]. In Section 2.3 the methodology is described by which the area of the fire was determined. The extent of the fire is indicated in Figure 3 in dark red.

2.2 Data Preparation

In this research, two different types of data are used. The Portuguese NFI (Uva et al., 2021), which provides information on general land cover in Portugal and specifically focuses on forests. Secondly, the Copernicus Sentinel-2 satellite images, which contains information about the pixels on spectral bands. These images are distributed as tiles with a size of 110 km x 110 km. The Copernicus Browser provides an overview of the Sentinel-2 images and allows filtering to relevant times and dates.

2.2.1 Tools and Workflow

The preparation and processing of the satellite images is done using the Sentinel Application Platform (SNAP). This is a tool developed by the European Space Agency. It is designed specifically for working with Sentinel-2 satellite images. It was used to create subsets of the

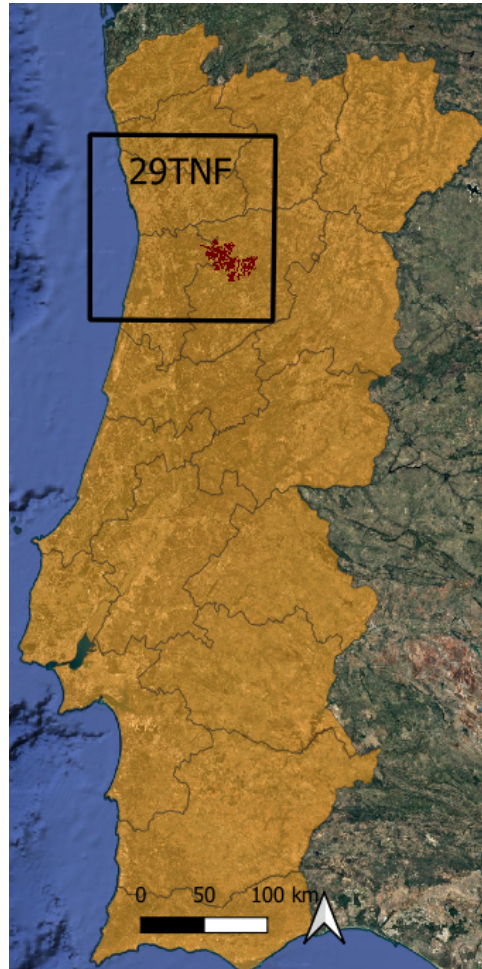


Figure 3: Tile 29TNF, Portugal With Fire Area (in Dark Red), Contains Modified Copernicus Sentinel Data [2024]

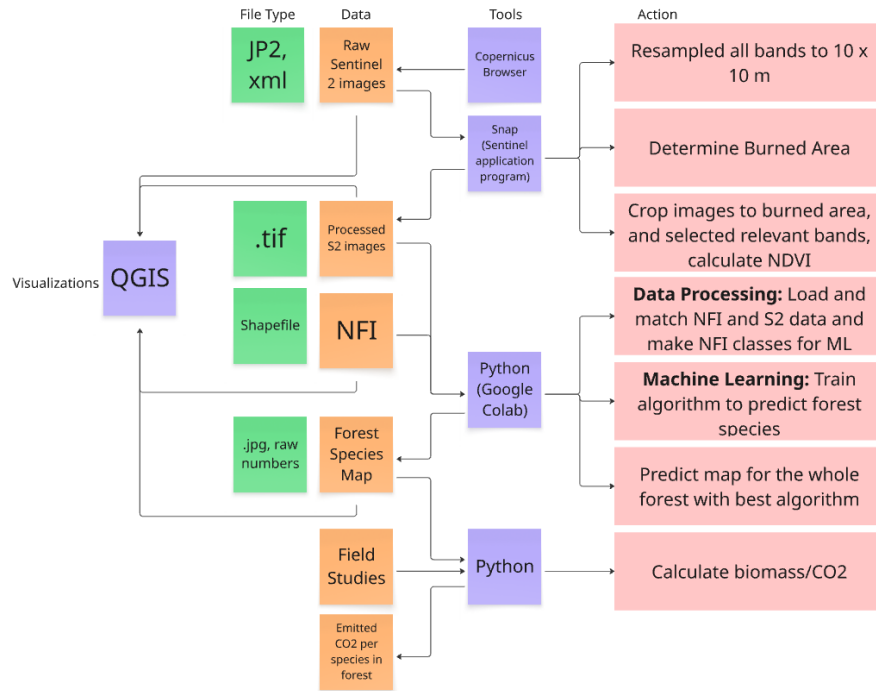


Figure 4: Tools and Data Pipeline: Data and Sources in Orange, Additional Information in Green, Tools in Blue and Actions in Red, With Additional Information in Yellow, Start From Top Middle

image that contained only the relevant area and the relevant bands. It was further used to calculate indices (NDVI, NBR) and to produce a mask of the burned area. The project was then exported as a GeoTiff file and imported in Python using the rasterio library. The second data source, the NFI, is loaded in Python using the geopandas library. Both libraries allow for the representation of different types of geospatial data in Python. The NFI is represented as specific points with a location (vector data), and the satellite image bands and indices are represented as continuous pixel grids (raster data). For analyzing and evaluating the NFI, data science methods are used. Machine learning is performed in Python in the Google Colab environment. Both data sources are matched based on their coordinate systems and loaded into one pandas data frame, which serves as the labeled feature space. Points outside of the mask are represented as "Not a Number" (NaN) and points without an NFI label are labeled as "unlabeled". The workflow of the project and the tools is visualized in Figure 4. The tools used for this project such as SNAP, Python and QGIS are open source tools and freely available.

2.2.2 Sentinel-2

The Sentinel-2 mission consists of two satellites, which give information about a certain area with an interval of 5 to 7 days (ESA, 2015). They provide information on 13 spectral bands, with a sensory range between 443 to 2190 nm. Four of the bands provide a resolution of 10 m, six bands have a resolution of 20 m, and three bands have a resolution of 60 m. Bands with

a resolution larger than 10 m were resampled to 10 m to ensure an even pixel size, following the examples of Moraes et al. (2025) and Ganz et al. (2020). The sampling method used was nearest neighbor. This method was selected to preserve the original spectral values of the lower-resolution bands. Hence, a pixel that previously had a resolution of 20 meters is split into 4 pixels with a resolution of 10 meters. They all inherit the value the larger pixel had before. This step is necessary in order to perform machine learning and to calculate indices, because the bands need to have the same resolution.

The bands: 2, 3, 4, 5, 6, 7, 8, 11 and 12 and the Normalized Difference Vegetation Index (NDVI) are used as features for the machine learning algorithm, as was done by Blickensdörfer et al. (2024) and Hemmerling et al. (2021), since they also worked on predicting the forest composition.

The NDVI is the ratio of NIR (Near Infra Red) light reflectance to Red light reflectance (Tucker, 1979). Hence, it is calculated with Equation 2, where NIR corresponds to Sentinel-2 band 8 and R to band 4.

$$\text{NDVI} = \frac{\text{NIR} - \text{R}}{\text{NIR} + \text{R}} \quad (2)$$

Table 3 gives an overview of the bands and the index that are used as features. Furthermore, it shows the resolution and the range of values of each of the features. The calculation of the NDVI was done in SNAP.

Features	nm	Description	Resolution	Range
Band 2	490	Blue	10 m	0–1
Band 3	560	Green	10 m	0–1
Band 4	665	Red	10 m	0–1
Band 5	705	Red Edge	20 m	0–1
Band 6	740		20 m	0–1
Band 7	783		20 m	0–1
Band 8	842		10 m	0–1
Band 11	1610	SWIR 1	20 m	0–1
Band 12	2190	SWIR 2	20 m	0–1
NDVI	x		10 m	(-1)–1

2.2.3 National Forest Inventory - IFN6

In this study, the most recent national forest inventory in Portugal, the IFN6 of 2015 (Uva et al., 2021), will be used. It provides detailed information on the forests in specific points. These points are spaced 500 meters apart. They contain information about the land cover in that area, and information such as forest size and density, if applicable. The categories it contains are "usosolo" (land use), "ocupprinci" (primary land cover), "ocupsecund" (secondary land cover), "tipo" (type of forest), "perccobert" (plant density) in percentage, "sobcoberto" (vegetation layer under land cover), and lastly, forest dimension in hectares. In this Section, the NFI points

within the study area are explored. In total, the study area contains 1,015 NFI points. The types of land use and their distribution can be seen in Table 4. Around half of the points indicate Shrubland, and slightly less than half of the points indicate Forest area. Some of the points indicate agricultural and unproductive / bare land.

Table 4: NFI Classes of *Land Use* in Study Area.

Land Use Class	Count
Shrubland	500
Forest	435
Unproductive/Bare Land	54
Agriculture	21
Inland Waters and Wetlands	4
Urban Areas	1

In Table 5 the distribution of primary and secondary land cover of the points where land use is *Forest* is depicted. It shows that most of the forest points indicate pure Maritime Pine stands and pure Eucalyptus stands. In addition, the region contains different types of mixed stands, such as Maritime Pines mixed with other trees. Table 6 visualizes how many times a tree species type occurs in primary or secondary land cover, in either class of land use. The forest inventory points give an indication of what kind of vegetation is present in the study area, but are not a continuous representation of the area. Additional information is needed to make statements such as "50% of the area is covered by Shrubland".

Table 5: Study Area Pairs of Primary and Secondary Land Cover in the Forest Land Use Class (With More Than 5 Occurrences)

Primary Use	Secondary Use	Count
Maritime Pine	Maritime Pine	189
Eucalyptus	Eucalyptus	133
Maritime Pine	Other Deciduous	28
Maritime Pine	Single Harvest	16
Other Deciduous	Other Deciduous	14
Maritime Pine	Eucalyptus	14
Maritime Pine	Shrubland	5

Table 6: Tree Species Occurrence in Primary or Secondary Land Cover in the Study Area

Tree Species	Occurrence Count
Maritime Pine	309
Eucalyptus	182
Other Deciduous	72
Oaks	11
Other Conifers	9
Chestnut	6

2.3 Identification of Burned Area

For this section, the SNAP tool was used. The satellite images before and after the fire were opened as separate products, and a subset with a smaller square was created that fully contains the area of the fire. This was done to reduce the computing load. For this area, the normalized burn ratio (NBR) was calculated before and after the fire with Equation 3 (as used in Arisanty et al. (2022)):

$$NBR = \frac{NIR - SWIR}{NIR + SWIR} \quad (3)$$

In Sentinel-2 images the band 8 corresponds to NIR light (Near Infrared), and band 12 corresponds to SWIR light (Short Wave Infrared).

The dNBR was calculated according to Equation 4, where the NBR after the fire is subtracted from the NBR before the fire.

$$dNBR = NBR_{before} - NBR_{after} \quad (4)$$

The dNBR indicates how much the NBR changed as an effect of the fire. The larger the difference, the more vegetation loss occurred during the fire. Hence, the NBR gives an indication of which area was burned in the fire. A threshold of $dNBR > 0.1$ was used to identify burned areas. This value was chosen according to the NBR classification table proposed by the United States Geological Survey (USGS), as cited in (UN-SPIDER Knowledge Portal, n.d.). A dNBR value of 0.1 indicates lightly burned areas, and higher values indicate more severely burned areas. For this research, 0.1 was used as the threshold, to capture the entire area of the fire, regardless of how severely it was burned. The dNBR with values ≤ 0.1 colored in black and values > 0.1 colored in white is visualized in Figure 5. The main body of the fire is clearly visible (in white), but there is significant noise around the fire. To create a smoother and less noisy image, the Erosion filter (5 x 5 pixels) was applied twice. Lastly, a polygon was mapped around the main area of the fire. Pixels outside of the polygon were removed. This was done to remove small patches of burned areas and noise outside the main area of the fire. This is visualized in Figure 6. This defines the exact area of the fire and the relevant pixels of the satellite images.

2.4 Forest Classification

In order to combine the high resolution of the satellite images with the low resolution high quality labels of the NFI, and to produce a high resolution continuous labeled map a machine learning approach was used. A Support Vector Machine (SVM) and a Random Forest (RF) classifier model were trained to predict the vegetation type for each pixel of the satellite image using information derived from the NFI as ground truth. The Sentinel-2 images were clipped to the extent of the mask of the burned area. This resulted in 2,526,593 pixels of 10 x 10 meters.

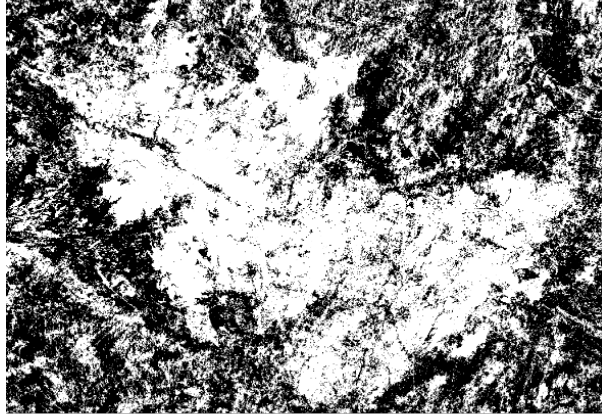


Figure 5: dNBR Colored With Threshold 0.1 (Black if Below Threshold, White if Above Threshold)

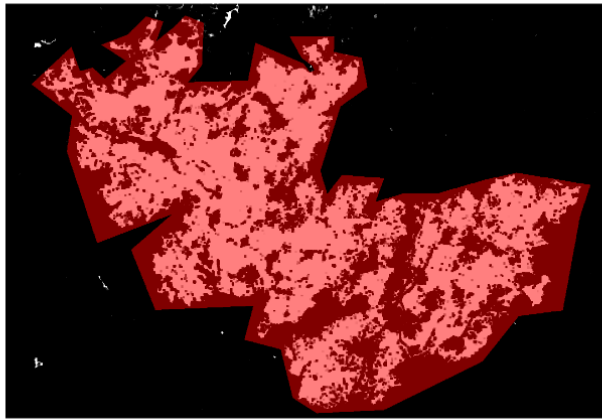


Figure 6: Filtered dNBR With Polygon Around Main Area

The NFI was also clipped to the extent of the mask (1,015 points). A 2D matrix was created that contained the 9 satellite bands (values between 0 and 1) and the NDVI (values between -1 and 1) as features. A column with labels was appended and initialized as "unlabeled". If available, the column was updated with a suitable label as described in the next section. For training the machine learning algorithm, all entries that were unlabeled were dropped and only the labeled entries were used.

Different types of models and model configurations were tested for the machine learning task. The best performing model was used to create a map of tree species over the whole fire area. This map has a resolution of 10 x 10 meters. The resolution is much higher than having one point of information every 500 meters as provided by the NFI.

2.4.1 Labels for Forest Composition Prediction

The NFI points were filtered to provide higher quality labels. NFI data processing for machine learning was performed in a similar manner as in Moraes et al. (2025). The 1,015 points in the

fire area were filtered based on:

- **Homogeneity:** The points where the primary land cover is the same as the secondary land cover
- **Size:** The points that indicate an area of at least 2 hectares

Filtering by homogeneity increases the chances that the NFI point indicates a homogeneous area that mainly contains the specified vegetation type. It is beneficial for the machine learning algorithm to learn from pure stands, to understand the associated spectral reflection patterns with fewer confounding factors. Furthermore, filtering by size allows increasing the number of labels generated from each NFI point. The application of the filters results in the distribution of land use and land cover that is visualized in Table 7. After filtering by homogeneity, the primary and secondary land cover types always match. Therefore, they will hereafter be referred to as land cover.

Table 7: NFI Labels After Filtering

Land Use	Land Cover	Count
Shrubland	Shrubland	271
Forest	Maritime Pine	157
Forest	Eucalyptus	92
Shrubland	Tall Shrubland	21
Forest	Other Deciduous	6
Agriculture	Dryland Crops	4
Shrubland	Burned Shrubland	3
Forest	Other Conifers	2
Unproductive Area	Unproductive Area	2
Agriculture	Irrigated Crops	1
Forest	Single harvest	1
Forest	Burned stands	1

After filtering, the points were further grouped to make suitable labels and reduce data complexity. The 5 labels are: (pure) Shrubland, (pure) Maritime Pines stands, (pure) Eucalyptus stands, other forest and non forest. The filtered NFI points were assigned to the classes as follows:

- **Eucalyptus:** The points where land use is *Forest* and land cover is *Eucalyptus*.
- **Maritime Pine:** The points where land use is *Forest* and land cover is *Maritime Pine*.
- **Shrubland:** All the points where land use is *Shrubland* and the land cover type is either: *Tall Shrubland*, *Burned Shrubland*, or *Shrubland*.
- **Other Forest:** The points where land use is *Forest* and land cover is either *Other Conifers*, *Other Deciduous*, *Single Harvest* or *Burned Stands*.
- **Non Forest:** The points where the land use is either *Agriculture* or *Unproductive Area*.

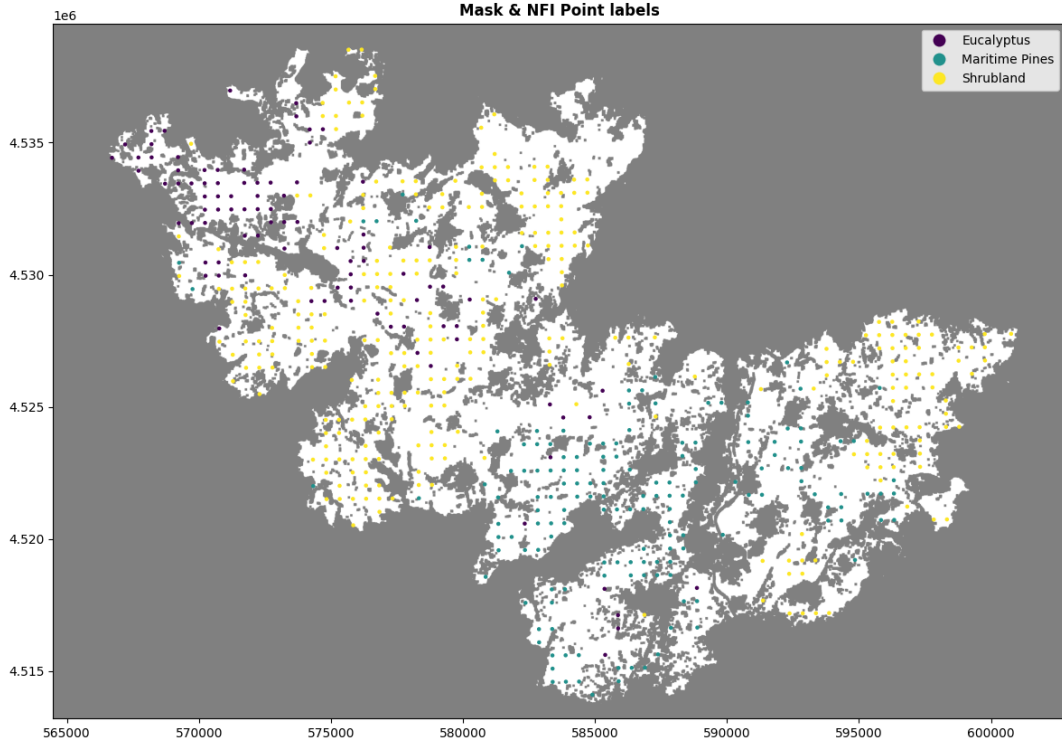


Figure 7: NFI Classes on Burned Forest Area After Label Preparation

The resulting class distribution is visualized in Table: 8. The classes "other forest" and "non forest" were dropped and not included as classes for the machine learning task. This was done because the goal of the work is to predict forest composition. The two classes are not indicative of a specific vegetation type. Moreover, the classes consist of only 10 and 7 instances, which indicate different land cover, and therefore express a high variance. For example, "non forest" consists of agriculture and unproductive area and "other forest" has other deciduous (6), other conifers (2), single harvest (1) and burned stands (1). The resulting task is a 3 class classification problem.

Table 8: Label Distribution: Forest Land Cover

Class	occurrence	label count
Shrubland	295	2,655
Maritime Pines	157	1,413
Eucalyptus	92	828
Other Forest	10	90
Non forest	7	63

Figure 7 visualizes how the NFI labels are distributed over the area of the fire.

After filtering and dropping the two irrelevant classes, 544 high quality points remained. By filtering by size, it was ensured that only larger areas of homogeneous forest stands were taken into account. Hence, each NFI point is considered as a true data point (label) for an area of 3 x 3 pixels, which corresponds to an area of 30 x 30 m. So each label in the NFI indicates 9 true

data labels. This is referred to as spatial augmentation. Of the 544 NFI points, 4,896 labeled data points are derived for training. Each NFI point is used to label the pixel that matches its location precisely and its 8 neighboring pixels.

2.4.2 Machine Learning

The data was split into train and test data with a distribution of 80% train and 20% test data. A standard scaler was applied first to the feature space in the training data and then to the test data. By labeling grids of 3 x 3 pixels and splitting them into test and training data, data leakage might occur, which could inflate the performance score of the algorithm. To prevent the algorithm from learning to recognize spatial clusters, rather than the pattern inherent to a certain land cover type, a group ID was assigned to each cluster of 9 labels derived from the same NFI point. A group split based on the group identifiers was used. Therefore, all 9 data points derived from an NFI point are either in the training or testing group and not split among the two. To test the impact of grouping the data, a fully random train test split method was used and their performance will be compared.

The model that was developed and is expected to perform the best used both filters, both satellite images and spatial augmentation. It will be referred to as the developed model. Different sets of hyperparameters were tested for the developed model. For SVM the RBF kernel was used and Bayesian optimization was performed in the range of $C = 0.1$ to 100 and $\gamma = 0.0001$ to 1. For Random Forest classifier the parameters in the range of n estimators = 10 to 250 and max depth = 1 to 30 were tested. The best combination was identified using the Bayesian Optimization library. The best parameters were identified for the developed model and applied to all other experimental configurations. This was done in order to reduce the complexity of the experiments and their evaluation. Tuning parameters for each experimental configuration would produce too much data and is not useful for the purpose of this experiment.

For evaluating the developed model, and for measuring and comparing the performance of the different models, a macro F1 score over all three classes was used, as they are equally important for the task.

The impact that NFI label processing has on the performance of the algorithm was tested. It was tested how filtering by both homogeneity and size affects the macro F1 score compared to not filtering the NFI data. Furthermore, it was tested how the performance changes when the NFI points are used to label only 1 point, or a window of three by three.

For testing the impact that using different satellite images as feature space has, it was tested how the algorithm performs if only the satellite image before the fire is used, if only the image after the fire is used, and if both of them (stacked horizontally) are used.

All the experiments with a 3 x 3 grid (spatial augmentation) were using the grouped train/test split method. In order to understand the impact that shuffling the data randomly without grouping would have, and whether this would lead to an inflated performance score, the developed model

was also tested with the regular train test split method, where all the points are shuffled randomly.

The general pipeline and the corresponding experiments are visualized in Figure 8. The impact of five factors on the performance of the machine learning models was measured: the choice of satellite images in the feature space, whether filters are applied, whether 1 or 9 labels are derived from each point of the NFI, whether a scaler is applied to the feature space, and whether the groups of 9 labels are kept together or randomly split between test and train data. All of the test configurations were tested against the developed model, while the other parameters were kept constant. The resulting experimental configurations are explicitly described in Table 9.

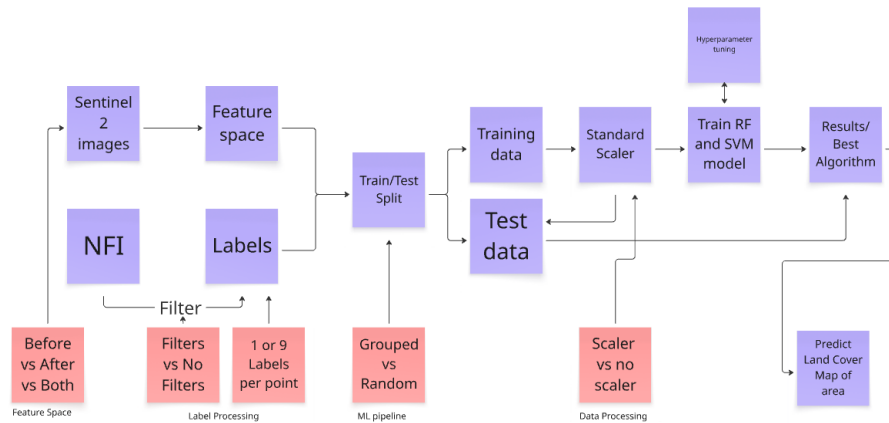


Figure 8: Machine Learning Pipeline With Experimental Comparisons in Red (From Left to Right)

Table 9: Experimental Configurations

Name	Satellite Images	Filters	Spatial Augmentation	Scaler	Test/Train
Developed Model	Both	Yes	Yes	Yes	Grouped
Before	Before	Yes	Yes	Yes	Grouped
After	After	Yes	Yes	Yes	Grouped
No Filters	Both	No	Yes	Yes	Grouped
1 x 1	Both	Yes	No	Yes	Grouped
No Scaler	Both	Yes	Yes	No	Grouped
No Grouping	Both	Yes	Yes	Yes	Random

2.4.3 Validation of Land Cover Map

To validate the accuracy of the predicted land cover map, it was compared to the Copernicus HRL Forest Type 2021 map (European Union's Copernicus Land Monitoring Service, 2021). This map indicates on a resolution of 10 x 10 meters, whether a pixel is non-forest, coniferous forest, or broadleaf forest. The pixels were directly compared to each other between the predicted

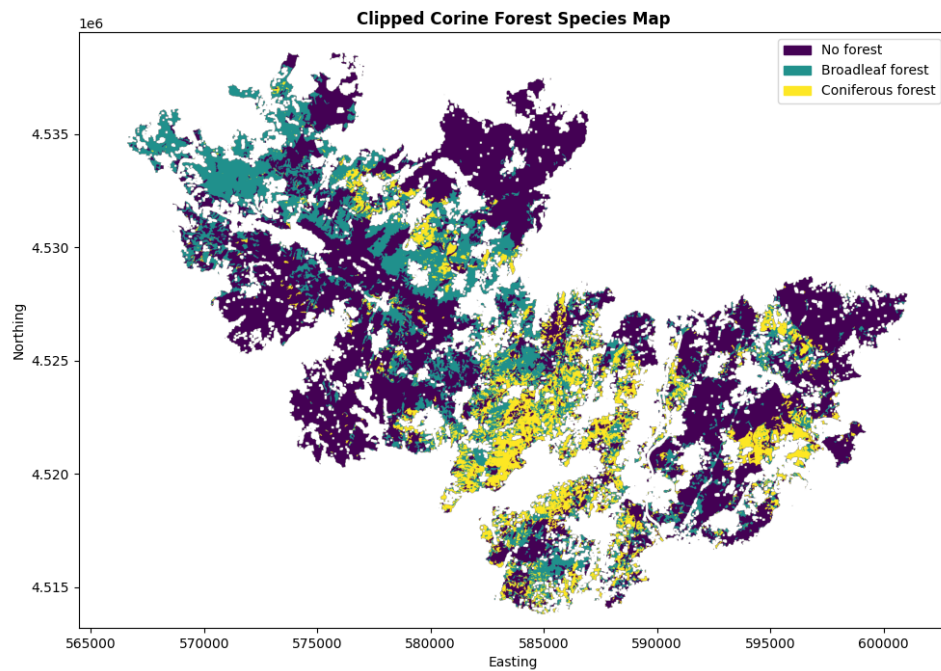


Figure 9: CORINE Map of Study Area for Validation

land cover map and the CORINE (Coordination of Information on the Environment) forest type map. The classes were compared against each other as follows:

- Maritime Pine was evaluated against the Copernicus Coniferous class.
- Eucalyptus was evaluated against the Copernicus Broadleaf class.
- Shrubland was evaluated against the Copernicus Non-forest class.

This grouping is possible due to the fact that Shrubland is considered to be non forest. Eucalyptus is the only broadleaf tree and Maritime Pines are the only coniferous trees predicted in this research. A spatial disagreement map was created to visualize where the two maps disagree. The validation map based on the CORINE data is visualized in Figure 9. In addition, the total forest distribution in the study area was compared, regardless of its location, between the CORINE map and the predicted land cover map.

Table 10: Biomass per Hectare for Shrubland, Eucalyptus and Maritime Pine Stands

Type	Biomass in Mg/ha	Reference	Remark
Maritime Pine	74.96 Mg/ha	ICNF (2019b)	46,100,000 Mg / 615,000 ha
Eucalyptus	78.70 Mg/ha	Alegria (2023)	Mean of their field study
Shrubland	16.40 Mg/ha	Alegria (2023)	Mean of their field study

2.5 Biomass and Carbon Emission Calculation

Usually, biomass estimates are based on (destructive) field samples. Collecting them was not feasible for this study. Instead, values from previous work based on field studies or the NFI were used. Vegetation biomass was estimated for each of the different vegetation types, in combination with a species specific value for biomass density. The area that a certain type of vegetation occupies (see Table 14) was multiplied by its vegetation density (see Table 10) to precisely determine how much biomass it contains. As a value for biomass density in Shrubland, the mean from the field study by Alegria (2023) in the north of Portugal was used. They found that Shrublands in their field samples had a mean height of 117.7 cm, a mean density of 43% and mean biomass per hectare of 16.4 Mg/ha. Alegria (2023) also calculated the mean biomass for Eucalyptus trees in the north of Portugal based on their field studies. In their 30 samples, they found that Eucalyptus stands had a minimum of 27.7 Mg/ha, a maximum of 169 Mg/ha and a mean of 78.7 Mg/ha. In this research, their mean value will be used. For the biomass of Maritime Pine tree stands, no local field studies could be acquired. Instead, an average based on the NFI biomass records was used. In the NFI6, it was reported, that Maritime Pines cover an area of 615 thousand hectares and have a biomass of 46.1 Mt (million tons) (ICNF, 2019a). Dividing the total biomass by the total area produces an estimate of the mean biomass density. The value is 74.96 Mg/ha.

Table 10 summarizes the values for biomass density, and where they originate from. The mean values for biomass in Mg per hectare were multiplied by the area in hectare that the corresponding vegetation types occupy to estimate the biomass in Mg. In this research, the biomass per stand type was estimated for a group of trees, rather than determining it on an individual tree level. This method used the vegetation distribution established in the previous step to estimate the emissions for each vegetation species individually.

2.5.1 CO₂ Equivalent of Total Biomass

The biomass in trees corresponds to the CO₂ they can release when burned. This relationship can be mathematically defined by a factor. There are different approaches to convert biomass to CO₂. The NFI gives a value for the biomass that the tree types contain and another value for the biomass equivalent in CO₂, if all the carbon stored in the trees combusts, binds to oxygen, and is released into the atmosphere. The factor of 1.834, as extracted as a mean from the Portuguese NFI (Section 1.4.5), was used to convert biomass to total CO₂ volume. This approach is not

suitable for describing CO₂ emissions caused by a fire, as it would lead to overestimations. The assumption, that all the available biomass in the trees would be released during the fire is not usually true. It is more realistic that only parts of the carbon in the above ground biomass of the trees are burned and converted to CO₂.

2.5.2 CO₂ Emission Calculation

Eggleston et al. (2006) formulated an equation to model the gasses released during a fire, capturing more details and complexity, and thus producing a more realistic result. Furthermore, they provided average values that describe how much of the total biomass of a specific type of tree is burned during the fire (combustion factor) and for how much CO₂ is released based on the biomass lost (emission factor). However, those parameters are not specific to Portuguese tree species, but rather global averages for tree species. Using this formula accounts for the typical burn behaviour of the individual species.

In this research the CO₂ emissions were calculated based on Equation 5 as proposed in the IPCC Guidelines for National Greenhouse Gas Inventories (Chapter 2) (Eggleston et al., 2006).

$$L_{\text{fire}} = A \times M_B \times C_f \times G_{ef} \times 10^{-3} \quad (5)$$

In this equation L_{fire} describes the amount of greenhouse gas, in this case CO₂. A is the burned area. M_B is the fuel density (biomass available per hectare). C_f is the combustion factor, and G_{ef} is the emission factor. The combustion factor expresses the proportion of available biomass that is burned during a fire. The emission factor expresses how many units of CO₂ are released per burned unit of biomass. This research estimates the area burned in the fire (A) for each of the vegetation species. For the combustion factor and emission factor, forest stand dependent values from Eggleston et al. (2006) are used (see Table 11). They expressed the emission factor in g/kg, which was divided by 1000 to produce kg/kg. This means that for each kilogram of Eucalyptus biomass burned, 1.569 kg of CO₂ are released into the atmosphere. By converting the emission factor to kg/kg, the value in the Equation is multiplied by 10^{-3} , so that 10^{-3} in Equation 5 cancels out.

Eggleston et al. (2006) pointed out that if the specific mass and combustion factor are not available, a default fuel value based on their research can be used (see Table 11). The value in parenthesis for fuel corresponds to the value specifically for wild fires, while the other value is the general fuel average. When available, the wild fire value was used for a better accuracy. For Shrubland no such value was provided in their research.

For the emission factor, a value specifically for Shrubland was used, and for both Maritime Pines and Eucalyptus stands, a mean for extra tropical forests was used.

For the combustion factor, no wildfire-specific values were available. For Eucalyptus stands the value for all Eucalyptus forests was used. For Shrubland, the value for Shrubland (general) has been used and for Maritime Pines, the value for all "other" temperate forests was used.

Table 11: Variables Associated With CO₂ Emissions for Different Stand Types, Extracted From Eggleston et al. (2006) (C_f = Combustion Factor, G_{ef} = Emission Factor)

Stand Type	Fuel $M_B \times C_f$	C_f	G_{ef}
Eucalyptus	69.4 (wildfire: 53) Mg/ha	0.63	x
Temperate Forest	50.4 (wildfire: 19.8) Mg/ha	0.45	x
Extra Tropical Forest	x	x	1.569 kg/kg
Shrubland	26.7 Mg/ha	0.95	1.613 kg/kg

The total CO₂ emissions per vegetation type were calculated as the product of the area burned (A) that is predicted by the machine learning algorithm, the biomass per hectare (M_B), that is inferred from the NFI (Uva et al., 2021) and from Alegria (2023), the Combustion factor C_f from Eggleston et al. (2006), and the emission factor G_{ef} from Eggleston et al. (2006). The final unit of this calculation is the weight in megagrams, which is equivalent to metric tons.

As a second approach, the fuel values from the IPCC report (Eggleston et al., 2006) were used instead of the biomass density estimates extracted from the NFI and Alegria (2023). For the fuel load approach Equation 6 is used.

$$L_{\text{fire}} = A \times \text{Fuel} \times G_{ef} \times 10^{-3} \quad (6)$$

It should be noted that for the fuel value (see Table 11) in Eucalyptus stands, a value specific to wildfires has been used. For Shrublands a general value has been used that is not specific to wildfires, because a value specific to wildfires was not available. For Maritime Pine stands the value for wildfire in temperate forests was used, which may introduce some inaccuracies, as temperate forests are a category that contains many other types of trees.

The two approaches are used, and the results will be compared. The first approach used local biomass estimates in Portugal, while the second approach used global fuel values (Table 11). Generally, the approach using local data will be considered to be more accurate and representative of the local vegetation.

3 Results

The results are split into three parts: the burned area detection, the machine learning algorithm for forest land cover prediction, and the biomass and CO₂ estimations.

3.1 Burned Area

The application of the dNBR threshold and subsequent filtering as described in Section 2.3 successfully delineated the fire area. According to this algorithm, 2.5 million Sentinel-2 pixels were part of the burned area. It was identified that the fire area consists of 25,265.93 hectares.

The result of identifying the burned area is visualized in Figure 10. This was used to crop both the NFI and the satellite images to the extent of the fire.

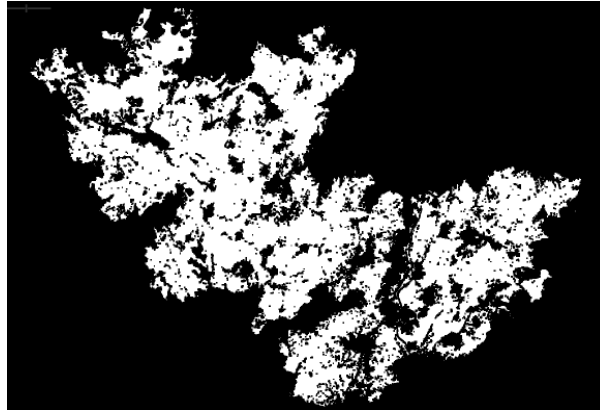


Figure 10: Mask of Burned Area

3.2 Machine Learning for Forest Land Cover Prediction

Performing hyperparameter tuning showed that for the developed model, the best parameters for the SVM classifier were $C = 18.4221$ and $\text{Gamma} = 0.0142$, and the best parameters for Random Forest were: $n \text{ estimators} = 180$, $\text{max depth} = 9$. The models with these parameters achieved an accuracy of 0.69 and a macro F1 score of 0.6 (SVM) and an accuracy score of 0.65 and a macro F1 score of 0.52 (RF). The F1 scores of the developed algorithm (SVM) for the different vegetation types are Shrubland: 0.82, Eucalyptus: 0.46 and Maritime Pines 0.53.

Table 12 gives an overview of the performances of the models with different configurations, as described in Section 2.4.2. This is also visualized in Figure 11. Table 13 gives a more detailed insight into the predictions of the developed algorithm using the image before the fire, after the fire, and both at the same time.

Table 12: Performance With Different Experimental Configurations

Setting	F1 macro (SVM)	F1 macro (RF)
Developed Model	0.60	0.52
Before	0.26	0.33
After	0.60	0.57
No Filters	0.49	0.48
1 x 1	0.54	0.49
No Scaler	0.41	0.52
No Grouping	0.70	0.73

A key finding is that algorithms that use both images and use only the image after the fire perform similarly well, whereas the algorithm that is trained only on the image before the forest fire performs much worse at the machine learning task. Table 13 further indicates that the model using only the image before collapsed into mainly predicting the majority class. The

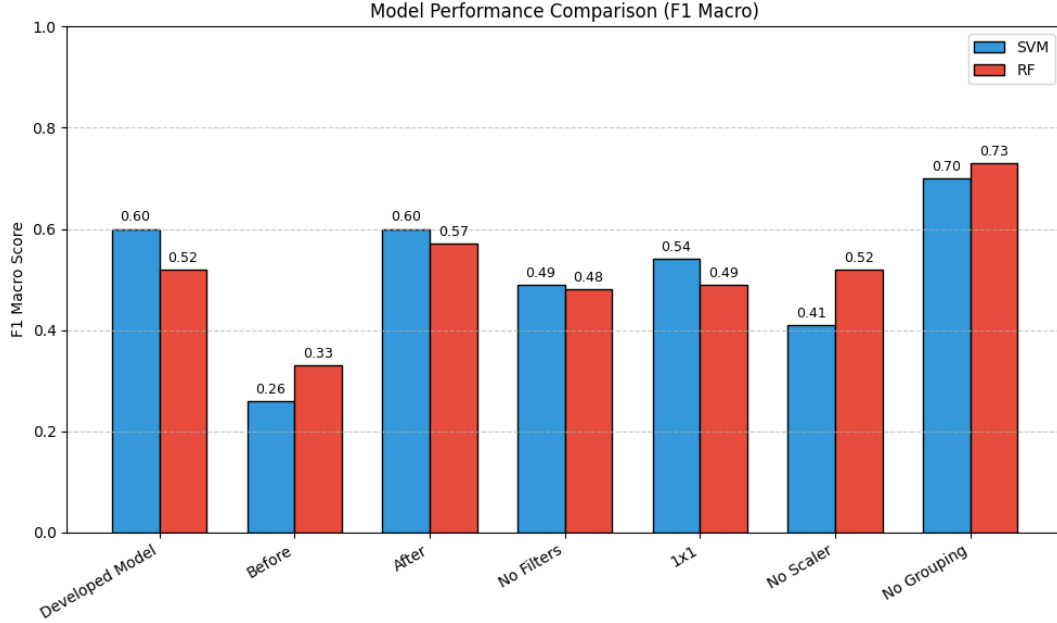


Figure 11: Comparison Between Different Configurations

other two models achieved a more robust separability between the majority class (Shrubland) and Maritime Pine stands. However, neither of them achieved good accuracy at distinguishing between Eucalyptus and Maritime Pine stands. To further investigate the potential cause of this, the band mean values and standard deviations have been plotted in dependence of the classes in the image before and after the fire. The visualization can be seen in Figure 13. It can be seen that bands 5, 6, 7 and 8 have a larger value on average in the image before the fire than after the fire. Moreover, in most bands the three classes appear to have a similar spectral reflection pattern. The NDVI is lower in the image after the fire, which is consistent with the fact that it indicates living vegetation. Moreover, the NDVI is much lower in the Shrubland class after the fire, than in the other two classes after the fire. Since Shrubland has a large combustion factor (most of the live biomass is burned in a fire), this is an expected outcome. The large difference in NDVI for Shrubland compared to the other classes may be the reason why algorithms that also use the image after the fire perform better at distinguishing between Shrubland and the two forest classes. The algorithm may be relying mainly on the NDVI to distinguish Shrubland from the other classes.

Comparing the difference between the developed model and the same setup without using filters shows that using filters improved the performance by 0.11 for the SVM and by 0.04 for the Random Forest classifier. Performing spatial augmentation improves the performance of the SVM classifier by 0.06 and of Random Forest by 0.03. This shows that both adding filters and labeling more points helped the algorithm learn the pattern behind the data better. This difference in both cases is stronger for the SVM classifier than for the RF classifier. The RF classifier seems to be more robust and less dependent on the processing of the labels.

Applying a standard scaler further improves the performance by 0.19 for SVM and does not

affect the performance of the Random Forest classifier. It can be seen that the SVM classifier is more sensitive to how the data is scaled. SVM is the algorithm that achieves the highest overall performance, but not applying a scaler worsens its performance, while Random Forest is not affected by it.

These experiments showed, that the steps in data preparation both for the feature space and the labels improve the performance of the machine learning algorithms.

Lastly, the configuration in which the test and train data were not split based on the groups but were fully randomly distributed achieved the highest macro F1 score. However, this does not indicate a better algorithm. The better performance is the result of the leakage of training data to the test data. It seems that the algorithm learned to identify the 3 x 3 based on the training pixels that were part of the grid. This underlines the great importance of keeping test and train data separate, when performing spatial augmentation. Otherwise, the performance score of the algorithm may not be indicative of how well it performs at the task it is designed for.

Using the best performing model, a prediction was made for all pixels of the burned area. Based on that, a predicted land cover map has been generated. The model predicted that 55.5% of the burned area are Shrubland, 31.57% are Eucalyptus stands and 12.93% are Maritime Pine stands. The map with the predicted classes is visualized in Figure 12.

Table 13: Confusion Matrix SVM and RF, First Row Indicating Eucalyptus, Second Row Maritime Pines and Third Row Shrubland (Column True Label, Row Predicted Label)

Feature Space	SVM	RF
Both, 3 x 3, both filters, grouped	$\begin{pmatrix} 70 & 65 & 58 \\ 28 & 127 & 80 \\ 15 & 49 & 469 \end{pmatrix}$	$\begin{pmatrix} 32 & 106 & 55 \\ 31 & 133 & 71 \\ 11 & 58 & 464 \end{pmatrix}$
After, 3 x 3, both filters, grouped	$\begin{pmatrix} 83 & 51 & 59 \\ 13 & 153 & 69 \\ 10 & 47 & 476 \end{pmatrix}$	$\begin{pmatrix} 42 & 110 & 41 \\ 14 & 148 & 73 \\ 1 & 66 & 466 \end{pmatrix}$
Before, 3 x 3, both filters, grouped	$\begin{pmatrix} 0 & 0 & 193 \\ 2 & 10 & 223 \\ 0 & 6 & 527 \end{pmatrix}$	$\begin{pmatrix} 5 & 24 & 164 \\ 12 & 81 & 142 \\ 26 & 139 & 368 \end{pmatrix}$

Table 14: Predicted Distribution of Vegetation Types Over the Fire Area

Vegetation type	Occurrence	Hectares	Relative (%)
Eucalyptus	797,676	7,976.76	31.60
Maritime Pine	326,647	3,266.47	12.93
Shrubland	1,402,270	14,022.70	55.47

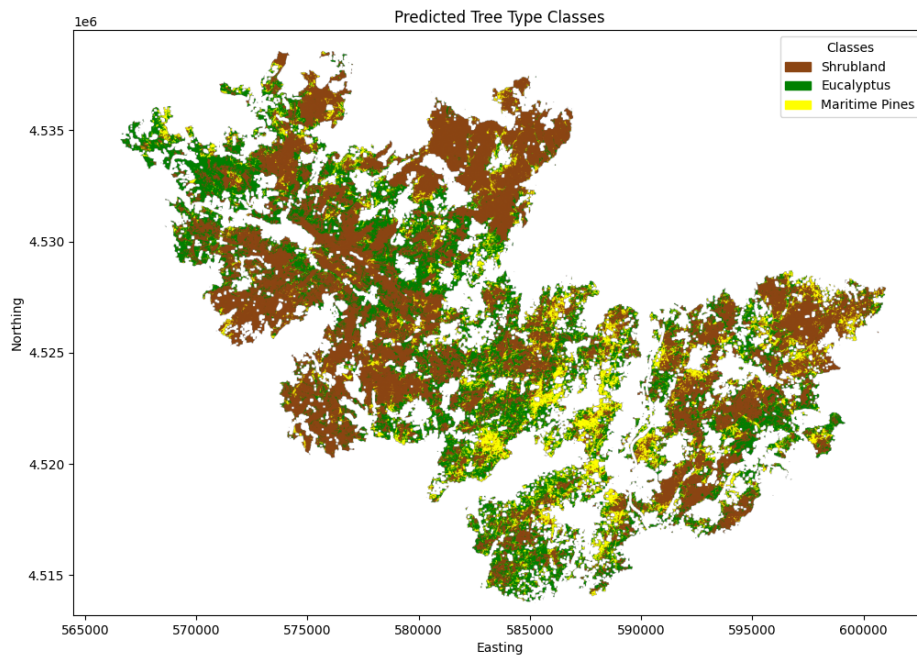


Figure 12: Predicted Vegetation Types Over Study Area

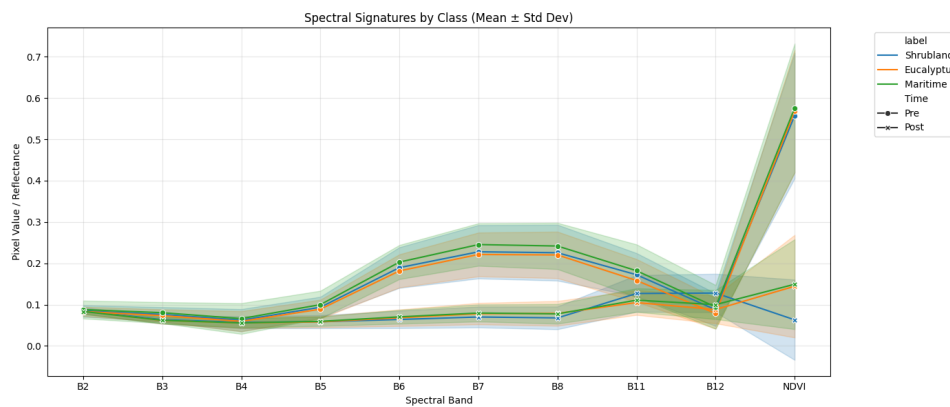


Figure 13: Spectral Analysis of Feature Space in Dependence of Class, Before and After the Fire

3.3 Burned Biomass and CO₂ Emissions

In this section, first the baseline prefire biomass is established. Subsequently, different methodologies are used to estimate the corresponding CO₂ emissions. The more detailed the method, the better it captures the complexity of the underlying patterns. The biomass per vegetation type is determined and based on that, their theoretical CO₂ capacity is calculated. Afterwards, the CO₂ emissions are calculated more precisely with a formula from IPCC, which takes into account species-specific factors.

3.3.1 Biomass Based on Area and Vegetation Stand Values

Multiplying the area that tree stands covered before the fire with the biomass per hectare they contained produces an estimate of the biomass present in the different stand types before the fire. The resulting biomass estimates are depicted in Table 15. This indicates that the Eucalyptus stands contained by far the most biomass. Furthermore, it indicates that Shrubland and Maritime Pine stands contain a similar amount of biomass, although Shrubland covers a much larger area. Shrubland is the most dominant vegetation species in terms of area, yet it contains significantly less wood volume per hectare than tree stands.

Table 15: Biomass Based on Alegria (2023), ICNF (2019b) per Land Cover Type

Type	Area (ha)	Biomass Mg/ha	Biomass (Mg)
Maritime Pine	3,266.47	74.96	244,854.591
Eucalyptus	7,976.76	78.70	627,771.012
Shrubland	14,022.70	16.40	229,972.280
Sum			1,102,597.883

3.3.2 CO₂ Capacity Based on NFI Factor

Using the biomass estimate from the last section and the factor estimated based on the NFI (Section 1.4.5) produces an estimate of the CO₂ equivalent of the biomass that tree stands in the study area contained. This is visualized in Table 16. However, this value is not an accurate representation of CO₂ released during the fire. It is not a reasonable assumption that all available biomass is burned during the fire. The value would be an overestimation of the actual CO₂ emissions, but it is a useful first approximation. The value is considered the maximum or upper bound of CO₂ emissions. The equivalent of total biomass in the area in CO₂ is 2,022,164.5 Mg (tons).

3.3.3 CO₂ Emission: Fuel Mass Approach

Using Equation 6 and the fuel load factor described in 2.5.2 for $M_B \times C_f$ produces the CO₂ emission results depicted in Table 17. This result is much lower than the CO₂ capacity that

Table 16: CO2 Emissions: NFI Factor and Associated Values			
Type	Biomass (Mg)	NFI factor	CO ₂ (Mg)
Maritime Pine	244,854.600	1.834	449,063.320
Eucalyptus	627,771.012	1.834	1,151,332.036
Shrubland	229,972.300	1.834	421,769.162
Sum	1,102,597.883		2,022,164.518

was estimated in the last step, because it accounts for the fact that not all biomass has been burned. Especially, this approach results in lower CO₂ emissions associated with Eucalyptus and Maritime Pine stands. Maritime Pine stands have a low fuel value because even though more biomass is present in Maritime Pine stands, not a large proportion of it usually burns. Eucalyptus has a relatively high fuel value, because it contains more biomass and also a large proportion of it usually burns.

The Shrubland values show an inverse trend. In the previous estimate it had 420 thousand Mg of CO₂ associated with it, and in this approach it increased to an emission of 600 thousand. This is due to the fact that the fuel load from IPCC is larger than the biomass per hectare measured by Alegria (2023), which was used in the previous step. Shrubland has a relatively low fuel value, which is due to the fact that it does not contain a lot of biomass, yet almost all of it burns in a fire.

This approach relies entirely on the default fuel load values provided by the IPCC for the different types of vegetation and does not take into consideration local values for biomass density.

Table 17: CO2 Emission: Fuel Mass Approach and Associated Values				
Type	Area (ha)	Fuel Load (Mg/ha)	G_{ef}	CO ₂ (Mg)
Maritime Pine	3,266.47	19.8	1.569	101,476.810
Eucalyptus	7,976.76	53.0	1.569	663,323.431
Shrubland	14,022.70	26.7	1.613	603,917.023
Sum				1,368,717.265

3.3.4 CO₂ Emissions: Mixed Approach

The third approach uses the locally derived biomass density and area and combines them with stand specific combustion and emission factors from the IPCC (Eggleston et al., 2006). This was calculated with Equation 5. Unlike the previous CO₂ estimation approach, this takes into account local data for tree biomass. This method yields a total emission estimate of 1.14 million Mg of CO₂. Since it uses both site-specific data and the a combustion factor, this figure is considered the most accurate representation of the carbon emissions caused by the fire in Reriz e Gafanhão.

Compared to the fuel load approach for measuring CO₂ emissions, the Shrubland emissions dropped significantly from 600,000 to 353,000. This is a result of the lower biomass density

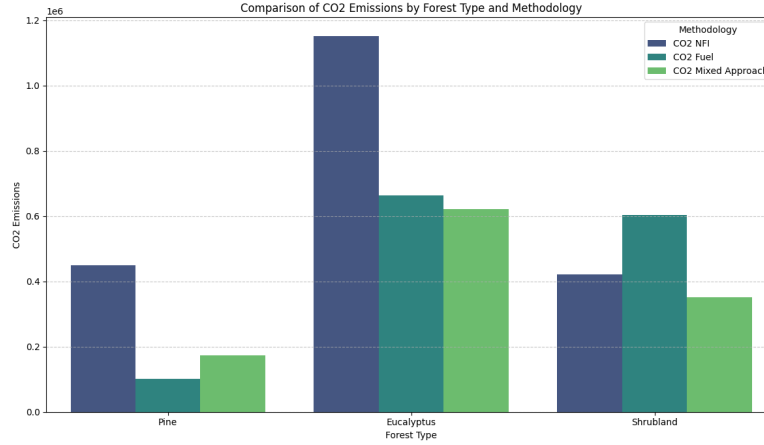


Figure 14: CO2 Emissions Calculated by Different Approaches (Capacity Blue, Fuel Approach Turquoise and Mixed Approach Green)

value measured in local field studies compared to the value from IPCC for Shrubland globally. The emissions caused by Eucalyptus stands decreased slightly compared to the last step and the emissions associated with Maritime Pine stands increased from 100,000 Mg to 173,000 Mg.

The different amounts of CO₂ associated with the vegetation types from the different equations are visualized in Figure 14.

Table 18: CO2 Emissions: Mixed Approach and Associated Values

Type	Area (ha)	M_B (Mg/ha)	C_f	G_{ef}	CO ₂ (Mg)
Maritime Pine	3,266.47	74.96	0.45	1.569	172,879.584
Eucalyptus	7,976.76	78.70	0.63	1.569	620,532.812
Shrubland	14,022.70	16.40	0.95	1.613	352,398.023
Sum					1,145,810.42

4 Validation

The CORINE land cover map is used to compare how many predicted pixels match the corresponding CORINE forest type map. The CORINE map indicates to which of the following classes a pixel of size 10 x 10 m belongs to: broadleaf, coniferous, or not forest. The CORINE classes are mapped one to one to the classes that were predicted in this research. The mismatch map (red for values that do not match between the two maps and green for pixels that match between them) is visualized in Figure 15. In total 1,613,985 pixels match between the two maps, and 912,608 pixels do not match. The confusion matrix is depicted in Table 19. The match accuracy is 0.639.

Treating the corresponding CORINE classes as true labels, the predictions of the best algorithm achieve an F1 score of 0.78 for shrubland, 0.54 for Eucalyptus stands and 0.28 for Maritime Pines stands. This indicates the same trend as observed in the test data. The

Table 19: Confusion Matrix: CORINE and Predicted Land Cover, CORINE Classes as Columns and Machine Learning Predictions as Rows

	Shrubland	Eucalyptus	Maritime Pine
Not Forest	1,109,889	208,473	114,923
Broadleaf Forest	196,948	407,274	114,902
Coniferous Forest	95,433	181,929	96,822

algorithm performs well at identifying Shrubland, but struggles to distinguish between the two forest classes. These misclassifications may be caused by the spectral similarity between Eucalyptus and Maritime Pine in the Sentinel-2 bands, making them difficult to separate based solely on spectral reflection patterns. Furthermore, it should be noted that the CORINE map is itself a predicted dataset rather than ground-truth field data. Therefore, the low F1 scores do not indicate that all of the predictions are strictly incorrect. Mixed stands of Eucalyptus and Maritime Pine trees might be classified differently in both maps, and a mismatch may occur, even though they are correct.

Lastly, the pixel distribution on the two images is compared. Their absolute count is visualized in Figure 16. For Shrubland, the relative error is 2.16%. For Eucalyptus stands, it is +10.92% and for Maritime Pine stands, it is -12.7%. In Figure 17 the relative distribution of the NFI, CORINE and the predicted map for the three classes is shown. It shows that the NFI distribution strongly deviates from the CORINE and the predicted map for the classes Eucalyptus and Maritime Pine. The NFI points contained 28.9% Maritime Pine trees, the CORINE map had 14.8% and the predicted map 12.9%. For Eucalyptus, a similar trend occurs with 16.9% of the NFI points being Eucalyptus stands and 31.6% and 28.5% of the predicted map and the CORINE map respectively being Eucalyptus stands. This shows that to some extent the algorithm was able to learn from the satellite image to produce a distribution that matches the CORINE map more closely.

5 Discussion

According to the EFFIS report (European Commission: Joint Research Centre, 2025), the fire used as a field study in this work burned an area of 35,523 hectares of forest. This research determined a fire area of 25,265.93 hectares. The map of the fire area, created in this study, has a spatial resolution of 10 meters and excludes human-made structures and other areas where no vegetation was burned, such as rivers, streets, and houses within the bounds of the fire area. In comparison, the report used Rapid Damage Assessment (RDA) (European Commission, 2026), which uses a semi-automatic classification procedure based on satellite imagery with a resolution of 250 meters and further refines it by incorporating Sentinel-2 images with a resolution of 20 meters. Hence, the difference in the estimated fire area may be caused by different resolutions and a different threshold or procedure for classifying the burned areas.

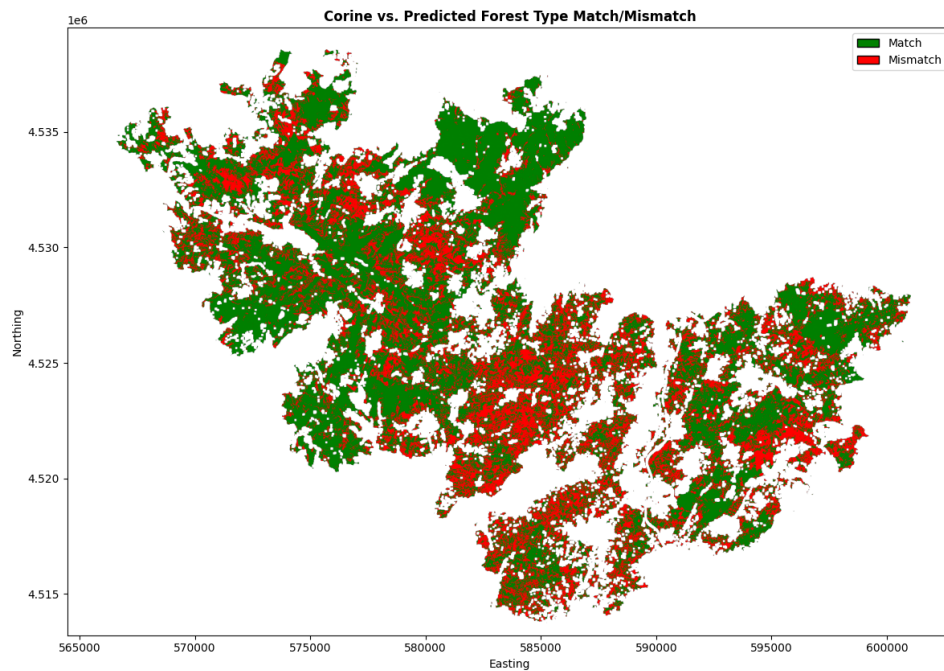


Figure 15: Mismatch Map Between Predicted Values and CORINE Map

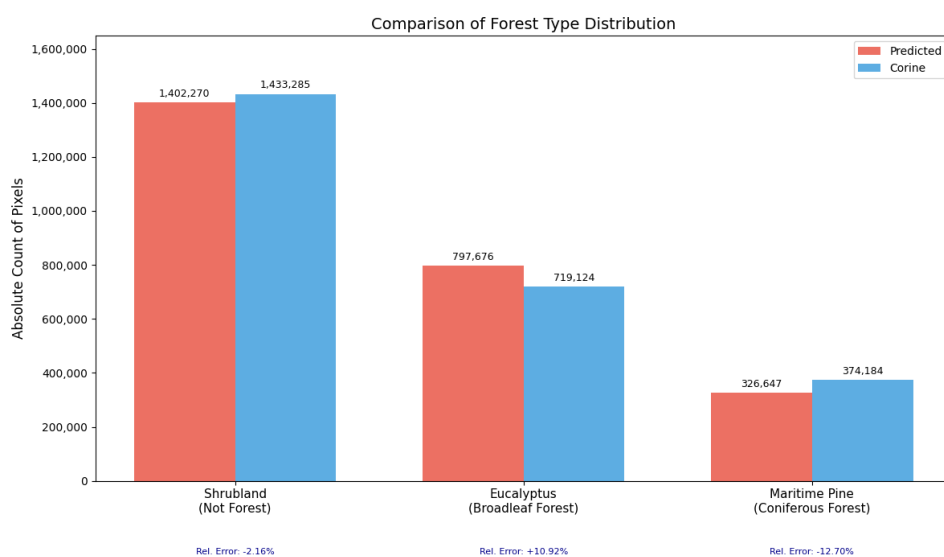


Figure 16: Vegetation Distribution: CORINE and Predicted Values

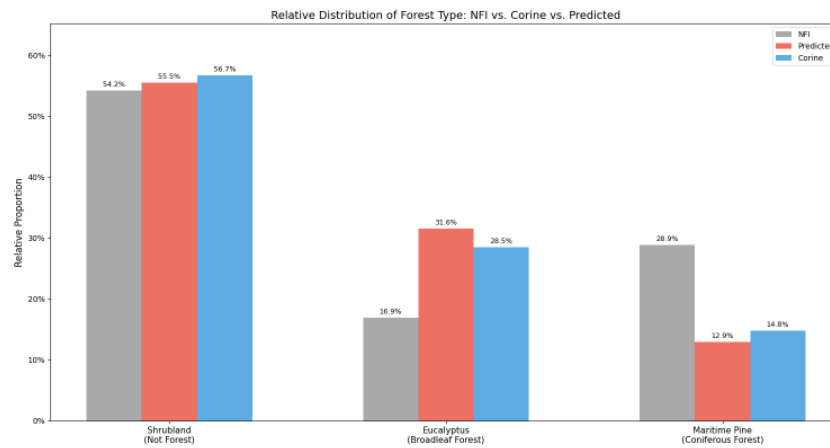


Figure 17: Relative Vegetation Distribution NFI, CORINE, and Predicted Values

The reliability of the final CO₂ and biomass estimates is fundamentally tied to the initial land cover classification, where the forest distribution was predicted. To address the sub-research question: *Using Sentinel-2 images and forest inventory data, to what extent can machine learning algorithms generate a more precise land cover map of the forest before the fire, when compared to using only forest inventory data?*, this research successfully showed that it is possible to make a more precise estimation of the forest distribution by combining Sentinel-2 images and NFI data. The correctness of the produced land cover map was assessed both by evaluating the machine learning performance and by validating it against an existing forest type map. The result provides a much higher resolution of information than the initially available forest inventory, yet there are certain areas in which the algorithm did not perform sufficiently well, as well as notable technical shortcomings that were not overcome. The primary limitation of the best land cover classification algorithm is that even though it performs well in distinguishing Shrubland from the two tree classes (F1 score of 0.82), it performs poorly in distinguishing between the two tree stands (F1 scores of 0.53 and 0.46). This may be caused by a similar spectral reflectance pattern of the two tree stands in the satellite images. Giving the algorithm more context in one or more of the following ways could help to overcome the classification challenges and increase the quality of the models predictions.

A common strategy to increase the performance of machine learning classifiers with geospatial data and forests is to make use of a time series. In the context of this research, using one image taken each month for the year before the fire would likely significantly improve the algorithm's performance. Using time series gives additional information to the classifier. Different trees look different and change differently throughout the seasons of a year. For example, coniferous trees typically do not lose their needles, while broadleaf trees usually lose their leaves in autumn. This is not the case for Eucalyptus trees, yet how they look through the seasons would still give more context to the algorithm. Adding a time series of images to the feature space is a straightforward method to increase performance for the classification task. The tradeoff is that more data processing would be required, increasing the size of the feature space, and

making algorithm training a more computationally complex task. Another way in which the performance of the model could be improved is by making use of more training data. Instead of only looking at the area of one fire, the same methodology could be applied to all the forest fires in Portugal for one year to significantly increase the number of training samples. This is expected to increase the performance, as currently only 5000 labeled instances were available. Another way to give more context to the classifier is to use the whole satellite image (or even multiple satellite images) and not just the fire area. This requires the processing of more data and makes the task more computationally complex, and was not done in this research, but would be useful for an engineering approach where the accuracy should be as large as possible. A third way to improve the performance of the algorithm would be to make use of convolution. Using convolution adds information about the neighboring pixels in addition to information about the pixel the prediction is made on. Since convolution gives more meaningful information to the algorithm, it is expected to improve its performance. The information is likely meaningful, as there is a strong correlation between the forest in one pixel and in the neighboring pixels.

Moreover, depending on the choice of algorithm, a potential technical challenge may arise from data processing in this work. By resampling with the "nearest neighbor" method, all 4 pixels of 10 x 10 m have the same value as the one 20 x 20 pixel before. This produces blocks, rather than a smooth landscape. For classifiers that treat each pixel independently, this is not as important as for classifiers such as CNNs, which aim to understand spatial patterns.

The results of the land cover classification further show that the algorithm performed well at distinguishing between forest stands and Shrubland, only when including the satellite image after the fire. Hence, including the satellite image after the fire adds important spatial information and drastically increases the accuracy of the machine learning model. This is a novel finding and potentially other research can benefit from including a satellite image after the fire when trying to perform land cover classification on a burned forest.

The forest land cover map can also be expressed as a forest distribution, which indicates how many hectares each vegetation type covers. This vegetation distribution serves as the basis of calculating the final CO₂ emissions caused by the fire. To address the main research question, *To what extent can a more precise estimation of a forest composition, using both forest inventory data and Sentinel-2 satellite images, enhance the accuracy of the estimation of CO₂ emissions produced by a forest fire?*, this research introduced a pipeline for more detailed mapping of the forest distribution and the calculation of associated CO₂ emissions. Using the forest distribution as a basis enables the use of more detailed and stand specific formulas and values, rather than averaging over the whole forest. The carbon emissions for each of the stands were calculated individually. This approach estimates fire emissions of around 1.1 million Mg of CO₂. However, the final estimate of CO₂ produced in this research could not be compared to ground truth data, because it is not available. Therefore, it is not possible to claim that the method introduced in this work produces a more accurate CO₂ estimate than previous approaches. Despite this, the methodology gives access to more spatial detail, providing a more granular view of the burned

area and accounts for vegetation specific factors in estimating the CO₂ released in the fire.

Additionally, the precision of this method for CO₂ estimation strongly depends on the quality of the field data. As a result, the accuracy of the final biomass and carbon estimation depends heavily on the quality — specifically accuracy, recency and locality — of the factors used in the equation. An insightful and necessary step in future work would be to validate this number by comparing it to real world measurements or by conducting field studies to validate or adjust parameters used in this formula. This would allow the methodology to be upscaled to larger areas and multiple fires. In this research, the biomass per hectare was inferred from local Portuguese studies from different sources. The fuel loads, combustion factors, and emission factors were not from studies specific to Portugal but instead come from a global average, as reported by the Intergovernmental Panel on Climate Change. Some of the values were not specific to the estimated stand type, but rather to the type of forest that this stand is a part of. This was the case for the emission factor in Eucalyptus and Maritime Pine stands. Similarly, for combustion factor and fuel load in Maritime Pine stands the value for Temperate Forest was used. All in all, forest distribution and biomass are specifically quantified for the different vegetation types, yet not all factors used for the calculation of biomass and carbon used local Portuguese data and stand specific values, due to the lack of their availability. In that regard, the research is limited by the data and does not overcome all of the challenges identified earlier. This study addresses the issue of inconsistencies due to averaging over large areas of mixed forests, by calculating the CO₂ for each species individually. This was done for the majority of the parameters, but not for all vegetation types, as some values were only available as averages for a broader vegetation class such as temperate forest. For Eucalyptus stands both a combustion factor and fuel load in wild fires were available. The value of CO₂ emissions caused by Eucalyptus stands burning may be more accurate than the one for Maritime Pines. In the future, the validity of the calculation of CO₂ emissions based on the forest distribution could be significantly improved by performing recent and local field studies and using the results, in combination with the proposed approach, to calculate biomass and CO₂ emissions based on forest distribution.

Besides the ecological data limitations that this work faced, it yielded several methodological insights. First of all, the results of the land cover classification part further underline the importance of keeping the training and test data strictly separate when performing spatial augmentation. If this step is left out in a machine learning project, the final accuracy may not at all represent how well the algorithm performs at the task and could lead to a high performance score even though the underlying pattern of the data is not understood.

Moreover, this work attempted using different strategies and algorithms that were discarded due to not being feasible. One strategy was to use machine learning to estimate the forest density as a percentage. The density predictions failed because the model was unable to learn meaningful patterns behind the data and the best accuracy that could be achieved was not better than random guessing. This research also attempted to use CNNs for the classification of forest land cover. However, the CNNs performed poorly on the classification task. As noted

previously, the cause may have been the blocky image created by resampling and the fact that there are sparse labels. CNNs usually rely on large amounts of annotated data and prefer a smooth image so they can detect patterns that might remain hidden with blocks. In order to use CNN algorithms, it would be necessary to generate a larger density and quantity of labels, and another resampling method should be attempted. A higher label density could be achieved by using weakly supervised methods (Moraes et al., 2025) or by having experts manually labeling part of the forest map; however this would be a costly and tedious approach.

The results of this work and the identified limitations give room for future work to explore. As another strategy to mitigate the blocky image introduced by resampling, future work could explore training two machine learning models: one on bands with a resolution of 10 m and one for the bands with a resolution of 20 m. An ensemble model could be used to predict the class based on the predictions of the two models.

Moreover, to leverage the ability of the dNBR to describe fire severity, it could be investigated to what extent it can be used as a predictor of how much of the available biomass in each pixel was burned. Because the dNBR gives an indication of the fire severity, it would be reasonable to investigate the correlation between the combustion factor (how much of the available biomass was burned) and the dNBR. If the correlation was high, the combustion factor per vegetation type could be scaled with the dNBR of each pixel. Calculating the dNBR is a straightforward, computationally inexpensive task. It could be measured if this approach performs better at capturing complexity in the burned area and adding important details to the calculations. This would further contribute to more detailed mapping of how the forests were burned and could potentially increase the accuracy of carbon emissions caused by forest fires.

6 Conclusion

The goal of this research was to develop a methodology for assessing the climate impact of forest fires in Portugal, specifically applied to the largest fire in 2024 in Reriz e Gafanhão. The primary objectives of this work were to investigate whether it is possible to create a high resolution map of the vegetation species in the burned area by combining satellite images with forest inventory data and to investigate whether this map can be used to calculate a more detailed estimate of the CO₂ emissions produced by the fire.

A 10 meters resolution map of the forest was successfully created by training a machine learning model using the forest inventory points as labels for the unlabeled pixels in the satellite image. The model was then applied to predict a value for each pixel in the fire area. This increased the density of information from one labeled point every 500 meters (discrete) to having every 10 x 10 pixel labeled (continuous). The best algorithm achieved a macro F1 score of 0.6, performing well at identifying Shrubland, but leaving room for improvement for the detection of tree stands. Despite the algorithm being prone to misclassifications, it greatly improves the density of available information. The algorithm predicted that before the fire 55%

of the forest area was Shrubland, 31% were Eucalyptus stands and 13% were Maritime Pine stands.

The detailed spatial distribution served as a basis for calculating the biomass and CO₂ emissions. The area that a vegetation type occupies was multiplied with species specific burn parameters to quantify the fire emissions. It was estimated that the prefire biomass in the area was around 1.1 million Mg. Converting biomass directly into its CO₂ equivalent leads to vast overestimates of around 2 million Mg. By taking into account the typical burn behavior and carbon contents of the different vegetation types, a more realistic estimate of around 1.14 million Mg of CO₂ was produced. Of those, the burning of Eucalyptus stands was estimated to have released 620,000 Mg of CO₂, the burning of Shrublands to have released 352,000 Mg of CO₂ and the burning of Maritime Pine stands to have released 173,000 Mg of CO₂.

In conclusion, a working pipeline for estimating the climate impact was introduced using only open source tools and publicly available datasets. Instead of averaging the CO₂ emissions over a whole forest, the pipeline produces a more granular map of the forest and allows for more detailed, species specific CO₂ calculations. In accordance with the values of IPCC and the NFI values for biomass, the CO₂ measurements are more precise, but whether the estimate corresponds to the real world emissions can not be concluded, due to the absence of local ground truth. A key methodological insight of this research is that a satellite image after the fire enhances land cover classification. This is a result that is easy to implement in future research in this field. In the future the validity of this approach could be further investigated by gathering field data and comparing the estimate to the real world CO₂ emissions. The same methodology can in principle be applied to any forest fire in Portugal to assess its climate impact.

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Summary

Forest fires in Portugal are a threat both to people and the environment. Due to climate change, the amount and severity of forest fires has been increasing in the last decades. Fire damage assessment is currently limited by relying on average values when working with large areas. A common strategy is to determine the size of a fire and multiply that with a constant for the type of forest such as mixed forest or broadleaf forest. This fails to account for species level variations in biomass and carbon emissions. This research aims to establish a methodology that produces a more detailed estimate of the climate impact of forest fires, that calculates the emissions on a species level. When calculating the emissions on a species level, values specific to the individual vegetation types can be leveraged for a more detailed description of the emissions.

The work is conducted specifically on the largest forest fire in 2024 in Portugal in Reriz e Gafanhão. Furthermore, it aims to present a reproducible pipeline that can be applied on a larger scale.

In order to calculate the emissions on a species level a detailed map of the burned area is necessary. The first step in achieving a map of the burned forest is to determine precisely the area of the fire. This was done using one Sentinel-2 image from before the fire and one from after the fire and calculating the dNBR, using a threshold point of 0.1 and applying filters to reduce the noise. The fire area was determined to have a size of around 25 thousand hectares. The study area in this research is precisely the area of the fire.

Satellite images provide the necessary resolution or density of information, but the pixels are not labeled and, therefore, the satellite image on its own is not suitable to describe the forest composition. The Portuguese forest inventory provides information about forests, but the information is sparse and only provides one point of information every 500 meters. The information is represented as discrete points. The density of those points is not high enough to infer statements about the forest distribution. This research aims to combine those two types of data to produce a higher resolution land cover map of the forest. This introduces the sub research question: *Using Sentinel-2 images and forest inventory data, to what extent can machine learning algorithms generate a more precise land cover map of the forest before the fire, when compared to using only forest inventory data?*

The combination of satellite images with the forest inventory and the application of a machine learning model yielded a 10 meter spatial resolution continuous map of the burned forest area. It was possible to generate a map with significantly more spatial detail than the forest inventory data. However, the approach also faced some technical limitations in regarding the performance of the algorithm. The limitations are related to the data preparation, data selection and algorithmic choices. The best algorithm achieved a macro F1 score of 0.6. The F1 score ranged from 0.82 for the majority class (Shrubland) to 0.53 (Maritime Pine) and 0.46 (Eucalyptus). Validating the created land cover map by comparing it to the CORINE forest

type map confirmed the findings, that the algorithm performs well at identifying the majority class (Shrubland) and often confuses the two tree classes. The performance of the algorithm could be improved by generally including more data, by using more satellite images as a time series, considering a larger study area with more labels and by adding convolutions in the form of information about the neighboring pixels. The final predicted map indicated that before the fire, 55% of the burned area was Shrubland, 31% was Eucalyptus, and 13% was Maritime Pine.

In the land cover classification it was further investigated whether the steps taken in data preparation actually improved the performance of the classification algorithm. Those steps included filtering the labels and performing spatial augmentation, scaling the feature space with a linear scaler and using as feature space two satellite images (one from before and one from after the fire). All of the processing steps improved the performance of the algorithms. A key finding of this research is that the satellite image after the fire adds information that allows the algorithm to make more meaningful predictions. The algorithm that only used the satellite image before the fire was not able to learn the pattern behind the data and ended up only predicting the majority class. By adding the satellite image after the fire, or only predicting based on the image after the fire, the algorithm performed significantly better.

Based on the area that the different species occupied before the forest fire, the research aimed to quantify the CO₂ emissions caused by each of the species individually. This introduces the main research question: *To what extent can a more precise estimation of a forest composition, using both forest inventory data and Sentinel-2 satellite images, enhance the accuracy of the estimation of CO₂ emissions produced by a forest fire?*

Multiplying the area that a tree species occupies with the biomass density gives an estimate of how much prefire biomass was present in trees of that species. From this the total biomass equivalent of CO₂ can be calculated, but it is not yet a realistic estimate of the emissions caused by the fire. Different types of vegetation contain a different amount of biomass per hectare, but also a different proportion of the available biomass burns on average, which is expressed by a combustion factor.

Applying the species specific fuel loads, combustion factors, and emission factors from the IPCC allowed for a more realistic estimate of impact of the fire. The research estimated that approximately 1.1 million Mg of biomass were present before the fire. It estimated that during the fire 1.14 million Mg of CO₂ were released into the atmosphere. The sum was attributed as follows among the three vegetation types: Eucalyptus stands 620,000 Mg, Shrublands 352,000 Mg, and Maritime Pines 173,000 Mg.

Using species specific parameters for the modeling of the CO₂ emissions produces a more detailed description of the emissions caused by a fire. This allows us to estimate how many percent of the total emissions may have been caused by the burning of each of the different species. This research question was answered only in parts. It was possible to conclude that this methodology accounts for variability on an individual species level and acknowledges that instead of treating the whole forest as a mixed forest the different species in the forest contain

different amounts of biomass, lose different amounts of biomass and even have a different ratio at which biomass is converted to CO₂. It was not possible to conclude that this approach produces a more accurate estimate, because there is no ground truth that this value can be compared to. Nevertheless, this research successfully developed a granular and detailed pipeline for assessing the climate impact of forest fires in Portugal, using open source tools and data, that overcomes limitations of broad averaging and which estimates CO₂ on a species level.